

The diversity of conjugative relaxases and its application in plasmid classification

María Pilar Garcillán-Barcia¹, María Victoria Francia² & Fernando de la Cruz¹

¹Departamento de Biología Molecular e Instituto de Biomedicina y Biotecnología de Cantabria (IBBTEC), Universidad de Cantabria-CSIC-IDICAN, Santander, Spain; and ²Servicio de Microbiología, Hospital Universitario Marqués de Valdecilla e Instituto de Formación e Investigación Marqués de Valdecilla (IFIMAV), Santander, Spain

Correspondence: Fernando de la Cruz, Departamento de Biología Molecular e Instituto de Biomedicina y Biotecnología de Cantabria (IBBTEC), Universidad de Cantabria-CSIC-IDICAN, C. Herrera Oria s/n, 39011 Santander, Spain. Tel.: +34 942201942; fax: +34 942201945; e-mail: delacruz@unican.es

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Abstract

Bacterial conjugation is an efficient and sophisticated mechanism of DNA transfer among bacteria. While mobilizable plasmids only encode a minimal MOB machinery that allows them to be transported by other plasmids, conjugative plasmids encode a complete set of transfer genes (MOB+T4SS). The only essential ingredient of the MOB machinery is the relaxase, the protein that initiates and terminates conjugative DNA processing. In this review we compared the sequences and properties of the relaxase proteins contained in gene sequence databases. Proteins were arranged in families and phylogenetic trees constructed from the family alignments. This allowed the classification of conjugative transfer systems in six MOB families: MOB_F, MOB_H, MOB_Q, MOB_C, MOB_P and MOB_V. The main characteristics of each family were reviewed. The phylogenetic relationships of the coupling proteins were also analysed and resulted in phylogenies congruent to those of the cognate relaxases. We propose that the sequences of plasmid relaxases can be used for plasmid classification. We hope our effort will provide researchers with a useful tool for further mining and analysing the plasmid universe both experimentally and *in silico*.

Introduction

Plasmid conjugation is a leading mechanism for genetic exchange in bacteria and thus an important component of bacterial evolution. It involves the cleavage of the transferring DNA in a site called *oriT* by a protein termed relaxase. As a result of the reaction, the relaxase becomes covalently bound to the *oriT* DNA. The resulting nucleoprotein complex is transported to the recipient cell by a protein export mechanism known as the type IV secretion system (T4SS). The DNA is actively pumped into the recipient cell by the type IV coupling protein (T4CP) (Llosa *et al.*, 2002; Christie, 2004). If a plasmid encodes the complete protein machinery for conjugal transfer, it is called self-transmissible or conjugative. Some plasmids contain a minimal gene set that allows them to be conjugally transmitted just when in the presence of a helper conjugative plasmid. They are called mobilizable plasmids. They usually contain just an *oriT*, a relaxase gene and one or more nicking-accessory proteins. Conjugative plasmids thus tend to be large (> 30 kb) with low copy number, while mobilizable plasmids are small

(< 15 kb) and have high copy number. We can generalize and say that all transmissible plasmids contain a MOB region, required for mobilization, while self-transmissible plasmids contain, on top of that, a T4SS that allows the assembly and functionality of the mating channel.

Relaxase proteins are large and usually contain two or more protein domains. The relaxase domain proper is located always at the N-terminus of the protein. At the C-terminus, a DNA helicase, DNA primase or other domain of unknown function is almost always found. In many cases (the exceptions will be noted in this work), relaxases contain a conspicuous signature (called 3H-motif) composed of a histidine triad that the protein uses to bind divalent cations. We already showed that relaxases are a convenient phylogenetic tool for plasmid classification, using a sample of mobilizable plasmids (Francia *et al.*, 2004). This tool has to be compared with classification via plasmid replication regions, carried out first by Southern hybridization (Couturier *et al.*, 1988) and later by PCR amplification (Carattoli *et al.*, 2005). Plasmid classification by replicon typing encounters problems derived from plasmids frequently carrying multiple replicons, mosaicism in replicons

(Boyd *et al.*, 1996; Osborn *et al.*, 2000), etc., as will be discussed in the fourth section. In this work, we proposed to complete the (Francia *et al.*, 2004) analysis by extending our survey to conjugative plasmids. This allowed the classification of conjugation systems in MOB families. The resulting families were MOB_B, MOB_H, MOB_Q, MOB_C, MOB_P and MOB_V. The most relevant features of each relaxase family were reviewed from the literature.

Method used for relaxase analysis

Relaxases are usually multidomain proteins, in which the relaxase domain always occupies the N-terminal position. Thus, PSI-BLAST (Altschul *et al.*, 1997) searches ($P=1e-4$ unless otherwise stated) were carried out using the N-terminal 300 amino acids of prototype relaxases from each MOB family as was done previously by Francia *et al.* (2004). Additional iterative BLAST searches using the phylogenetically most distant family members were carried out in order to cover a maximum sequence space without corruption. Data from relaxases sequenced up to December 2007 were included. BLAST searches organized the relaxase universe in specific protein families. This resulted in well-defined family boundaries in all cases except in the MOB_P family. In this case, when some deep branching proteins were used in iterative BLAST searches, they could retrieve additional members. However, it was considered too speculative to include data from such remotely related proteins for which no additional functional information was available. Thus, the MOB_P family should be presently considered as unfinished. Parallel BLAST searches with T4CP sequences (generally adjacent to relaxases in genetic maps) were used as confirmation for the presence of a functional conjugative system. Only relaxases contained in plasmids or in demonstrated integrative and conjugative/mobilizable elements (ICEs/IMEs) with a T4CP gene in the genetic vicinity were included in the present phylogenetic and molecular analysis. The resulting tables of constituent plasmids and ICEs/IMEs are presented as supplementary information. Multiple alignments were carried out using CLUSTALW (Thompson *et al.*, 1994). The phylogenies were constructed with MEGA version 3.1 (Kumar *et al.*, 2004). For each MOB family, we calculated a phylogenetic tree using neighbour-joining (NJ) analysis, tested with bootstrap values (1000 replicates). Topologies were confirmed by maximum likelihood (ML) analysis – not shown in the manuscript – using PHYML (<http://atgc.lirmm.fr/phyml/>) (Guindon & Gascuel, 2003; Guindon *et al.*, 2005).

Classification of large conjugative plasmids

Francia *et al.* (2004) classified relaxases of small mobilizable plasmids in four main families or superfamilies. To attempt

the classification of large conjugative plasmids, we conducted a database and literature analysis using the amino acid sequence similarity of their relaxases as a first criterion. In general terms, the most important difference with respect to the survey of mobilizable plasmids of Francia *et al.* (2004) is the addition of two new relaxase families that are present predominantly in large conjugative plasmids: the MOB_F and the MOB_H families. They will be analysed first (see The MOB_F family and The MOB_H family, respectively) because they are well-resolved families in phylogenetic terms. The MOB_C family is quantitatively smaller but has grown significantly since the previous review (it was then called CloDF13 family and contained only three plasmids). Because MOB_C contains a clade of conjugative plasmids, it merits a specific discussion (see The MOB_C family). Besides these three families, the remaining three families showed less well-defined contours and perhaps all three form part of a much larger protein class (hence, we will call it the MOB_P cluster). First, the MOB_P superfamily (see The MOB_P cluster) is greatly enlarged; it now includes the MOB_{HEN} family within it. It also contains several branches that are conspicuous but contain too few sequences to be properly analysed. The MOB_Q family is also enlarged and partially overlaps MOB_P (see The MOB_Q family). Finally, the MOB_V superfamily also grows, is very diverse, and overlaps MOB_P, and, in our opinion, remains generally underanalyzed because of a relative paucity of sequences. In summary, we will describe six relaxase families or superfamilies (see The MOB_V cluster). Because some of the relaxase families have already been discussed in length in previous publications (Francia *et al.*, 2004; Fernandez-Lopez *et al.*, 2006), they will just be updated here specifically in the contribution of large conjugative plasmids.

To our knowledge, all conjugative systems represented in the DNA sequence databases (> 1000 in December 2007) contain relaxases that are contained in one or another of these six families, with the exception of Tn916 (see Other relaxases: Tn916 relaxase). A schematic representation of relaxase diversity is shown in Fig. 1. A total of 616 plasmid relaxase sequences were analysed. The figure represents the different relaxase MOB families by circles whose sizes represent their quantitative representation in databases. Circles overlap if there are members that belong to more than one group (because they appear in BLAST searches started from two different prototypes).

It should be noted that in many examples shown in this work, terminal branches are compressed in the phylogenies. They represent several or even a large number of independent isolates, frequently containing plasmids with the same backbone but different cargo (i.e. antibiotic resistance genes). See, for instance, the F/R1 type of IncF plasmids (Boyd *et al.*, 1996), the IncP β plasmids (Heuer *et al.*, 2004) or the IncW plasmids (Revilla *et al.*, 2008). Thus, each

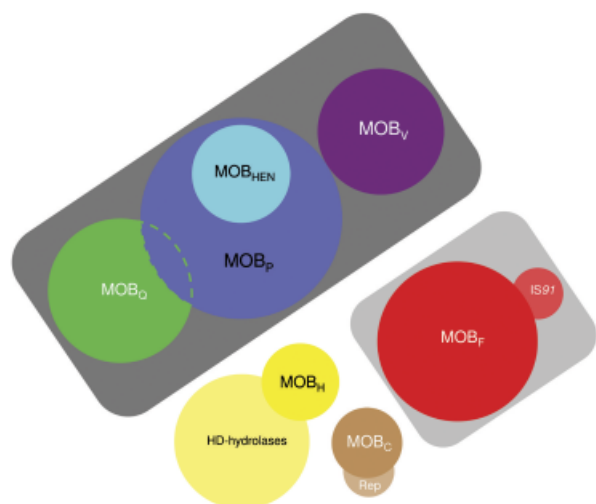


Fig. 1. A scheme of the relationships between the main relaxase protein families. A first relaxase cluster (shown on a dark-grey background) contains relaxase groups that contain just one active Tyr in the catalytic centre (see text). A second relaxase cluster (light-grey) contains relaxases with two Tyr in the catalytic centre. For the remaining groups (white background) not enough is known about the biochemistry of the respective relaxases. According to the analysis of their amino acid sequences and of experimental work (see text) they seem to be nonhomologous, and thus may use different DNA-processing mechanisms. Some relaxase protein families also overlap other protein families, such as plasmid RC-replication proteins (Rep), IS91-like transposases (IS91) or HD hydrolases. Areas of circles are proportional to relaxase number. The MOB_P area includes (MOB_P + MOB_{HEN} + MOB_{Q1}).

terminal branch in our phylogenetic trees names a prototype that may represent several relaxases with even < 95% amino acid sequence identity.

Main families of conjugative plasmids

The MOB_F family

This is a well-analysed family, which contains 114 members (Supporting Information, Table S1). For two of their relaxases, there is detailed structural and biochemical knowledge: TraI_F (Larkin *et al.*, 2005; Hekman *et al.*, 2008) and TrwC_{R388} (Guasch *et al.*, 2003; Gonzalez-Perez *et al.*, 2007). The relaxases of the MOB_F family are large proteins that consist of two domains: an N-terminal relaxase domain in which there are two catalytically active tyrosines and a C-terminal helicase domain. The helicase domain is characteristic of MOB_F relaxases and probably constitutes an important adaptation cue of these conjugative systems. The phylogenetic origin of the helicase domain in MOB_F relaxases has already been discussed (Fernandez-Lopez *et al.*, 2006). Besides these two domains, TraI_F was reported to contain a third domain in its C-terminus, implicated in

conjugative transfer (Matson & Ragonese, 2005). Similarly, TrwC contains a third domain between the relaxase and helicase domains (Cesar *et al.*, 2006). MOB_F relaxases contain two catalytic tyrosines in their catalytic centre (Grandoso *et al.*, 2000). This is another important peculiarity of this relaxase family and can have consequences in the mechanism of transfer (Gonzalez-Perez *et al.*, 2007). Also, MOB_F *oriTs* are comparatively large and complex, presumably reflecting highly regulated DNA sequences. The diversity and long-term evolution trends of the MOB_F family were already discussed by Fernandez-Lopez *et al.* (2006) and the reader is referred to this paper for an exhaustive analysis of the genetic assembly and evolution of the MOB_F genetic determinant.

Figure 2 updates the phylogenetic tree of the family. The PSI-BLAST search was carried out using the threshold value $P = 1e-8$ and converged in the fourth iteration. Using a lower threshold, such as the standard $P = 1e-4$, retrieved transposases of the IS91 family (Garcillan-Barcia & de la Cruz, 2002). This fact is interesting in itself and suggests that MOB_F relaxases and rolling-circle transposases are phylogenetically related, albeit remotely. As observed before, the crystal structure of the MOB_F relaxase TrwC_{R388} clearly shows that it is a linear permutation of the structure of rolling-circle (RC) replication initiator proteins, to which IS91 belongs (Garcillan-Barcia *et al.*, 2002; Guasch *et al.*, 2003).

As can be observed in Fig. 2, there are several relatively well-resolved clades in the MOB_F family. Clade MOB_{F1} is, for the time being, the most numerous, and includes plasmids from *Proteobacteria* and *Cyanobacteria*. Other clades include relaxases of the phyla *Proteobacteria* and *Actinobacteria*. They are not so well resolved at present, probably because an insufficient number of members have been analysed. The MOB_{F1} clade is of the utmost importance in *Gammaproteobacteria* and includes plasmids of the classical IncF, IncN, IncP9 and IncW incompatibility groups. Three differentiated subclades within MOB_{F1} – also supported by ML analysis – are MOB_{F11}, MOB_{F12} and MOB_{F13}, which are coloured in Fig. 2. The MOB_{F11} subclade contains two defined branches, represented by the prototype plasmids R388 (IncW) and R46 (IncN) for the enterobacterial branch, and by pWWO (IncP9) for the *Pseudomonas* branch. IncN plasmids are involved in the dissemination of multidrug resistances in *Enterobacteria* (Shen *et al.*, 2008). IncP9 plasmids are involved in the dissemination of catabolic pathways for xenobiotics (Greated *et al.*, 2002). In all MOB_{F11} plasmids, relaxase and T4CP genes are adjacent in the DNA sequence. Subclade MOB_{F12} is comprised by the phylogenetically broad IncF complex. It is as broad, in phylogenetic terms, as the MOB_{F11} clade. It should be noted, however, that the IncF complex was defined not by incompatibility, but by sensitivity to F-specific phages. In fact, the

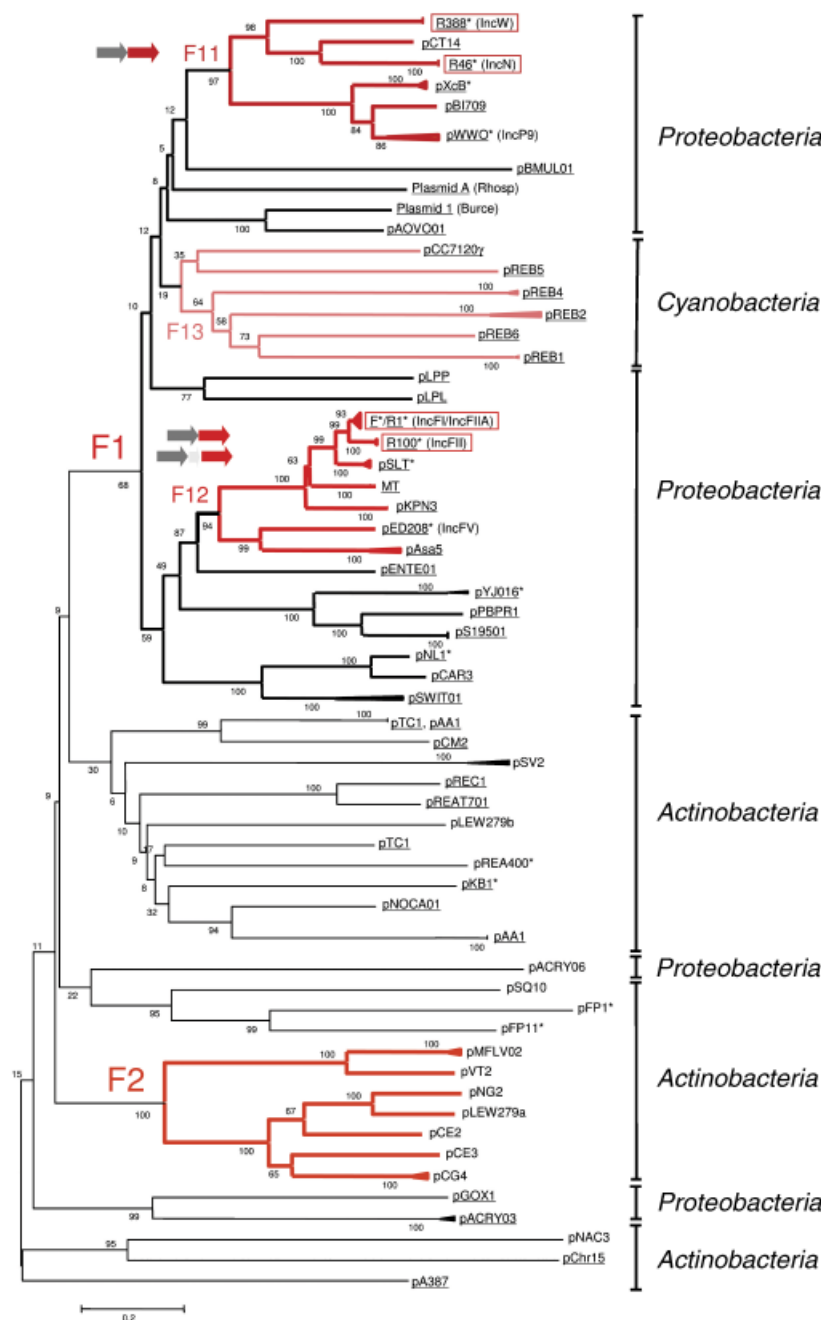


Fig. 2. Phylogeny tree of MOB_F relaxases. The dendrogram was constructed using a NJ algorithm. Bootstrap values for 1000 replicates are indicated. The tree was rooted with TrwC_R388 most distantly related homologues (pA387, pChr15 and pNAC3) that still belong to MOB_F. The distribution of plasmid origins according to the bacterial phyla is shown by vertical bars. Some specific clades that are discussed in the text are emphasized in colour with different tones of red and labelled F1, F2, etc. Plasmids containing a T4CP gene in the vicinity of the relaxase gene (and thus presumed to be conjugative) are underlined. In the clades where the synteny relaxase-T4CP was conserved, this is shown in the figure by thick horizontal arrows where the T4CP is coloured dark grey, the relaxase gene is coloured red and any intervening gene(s), light grey or a thin black line. Plasmids whose relaxases have been analysed biochemically are boxed. Plasmids for which conjugation has been experimentally demonstrated are labelled with an asterisk. Rhosp, *Rhodobacter sphaeroides*; Burce, *Burkholderia cenocepacia*.

IncF complex includes at least seven incompatibility groups (de la Cruz *et al.*, 1979). IncF plasmids are prevalent in *Enterobacteria*, and frequently responsible for the dissemination of antibiotic resistance determinants of medical importance (see e.g. Womble & Rownd, 1988; Coque *et al.*, 2008) and virulence factors (Rotger & Casadesus, 1999; Herrero *et al.*, 2008). Along with the MOB_{F11} and MOB_{F12} clades, a possible subclade MOB_{F13} includes a series of plasmids from *Cyanobacteria*. When more sequences become available, this subclade will probably further subdi-

vide; thus, its status is tentative. In any case, it seems clear that plasmid relaxases group in phylum-specific subclades, a result that will appear consistently in this review.

Besides clade MOB_{F1}, the only other clade that is emphasized in the figure is MOB_{F2}, which is well supported by NJ and ML analyses. It groups a set of plasmids of the phylum *Actinobacteria*. This is the only group within MOB_F, together with the *Streptomyces* linear plasmids represented by pSV2, which do not contain an associated T4CP (and are thus presumed to be mobilizable, not conjugative). Besides

the most conspicuous subclades that are numbered in Fig. 2, there are other scattered members in poorly resolved branches. Because of the low overall amino acid sequence conservation, resolution of the deep branches is not reliable. In general, this will be the case in all trees presented in this work. Thus it is not possible to present a complete or final plasmid classification, which will need many more sequences to be analysed.

The conserved signatures in the best-analysed clades of the MOB_F family are shown in Fig. 3. Sequence conservation is extensive in the three relaxase motifs previously shown to configure the protein catalytic centre (Guasch *et al.*, 2003; Francia *et al.*, 2004). In TrwC, Y18 and Y26 are the catalytic nucleophiles, which attack the scissile DNA phosphodiester bonds. D85 abstracts a proton from Y18 (and presumably also from Y26) and thus helps to configure Y18 as a potent nucleophile. The histidine triad (H150, H161 and H163) is involved in coordination of a Mg²⁺ ion, essential in the cleavage mechanism. Considering the 114 relaxase proteins

in the complete family, motif III is the most conserved. H161 and H163 are conserved in all proteins (the only two invariant amino acids). H150 is present in all proteins except in TraI_pLPL, where it is substituted by a Q. Within motif II, D85 is also highly conserved, with only three exceptions: in two cases it is changed to an E, and in the third (plasmid pACRY06) the alignment is uncertain. Motif I is more difficult to scrutinize, because there is smaller overall conservation. A nonexhaustive survey suggests that both tyrosines are invariant.

The phylogeny of the MOB_F T4CPs is compared with that of the MOB_F relaxases in Fig. 4. As can be seen, both phylogenies are highly congruent. The branches in the F11, F12 or F13 clades are almost completely conserved. Only the positions of the most profound clades vary with respect to one another. The same congruence is observed in ML topologies. This result indicates that relaxase and T4CP evolve together, a fact that was evident also in the analysis of other relaxase families (data not shown).

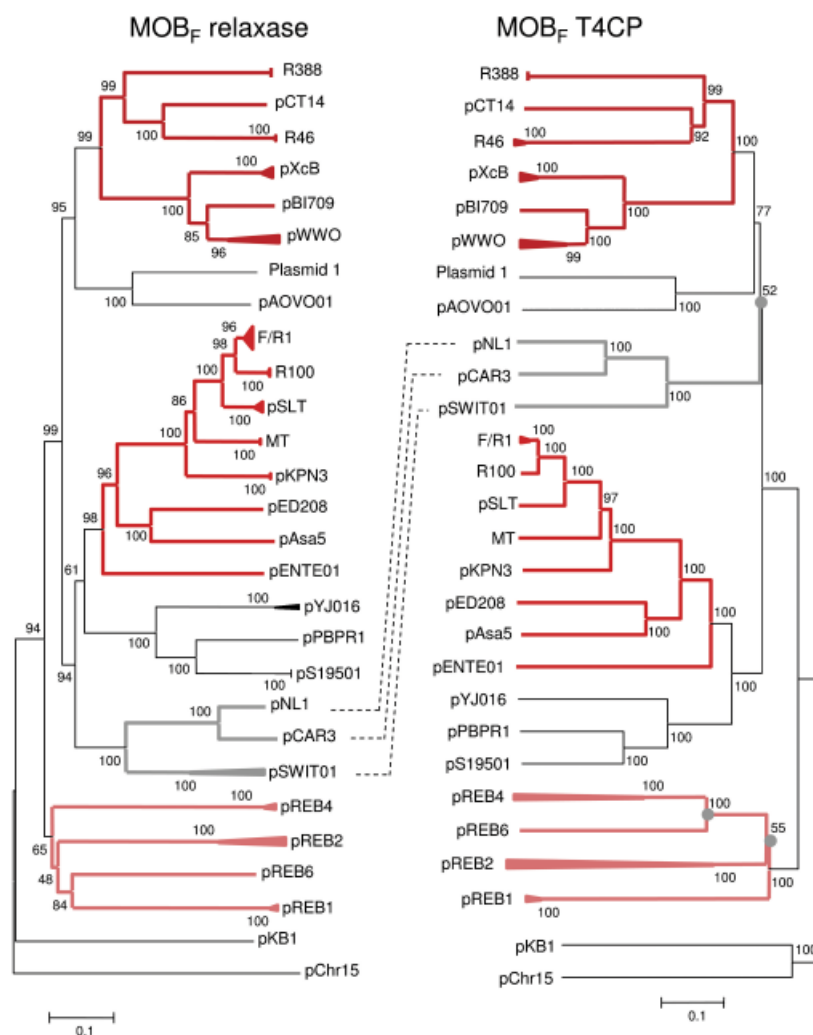


Fig. 3. Comparison of MOB_F relaxase and T4CP phylogenies. Branches of the main clades F11, F12 and F13 are coloured in different tones of red as in Fig. 2. Dotted lines indicate a swapped clade, whose branches are grey coloured. A grey circle is placed in the nodes of the T4CP tree that are not congruent with the relaxase phylogeny.

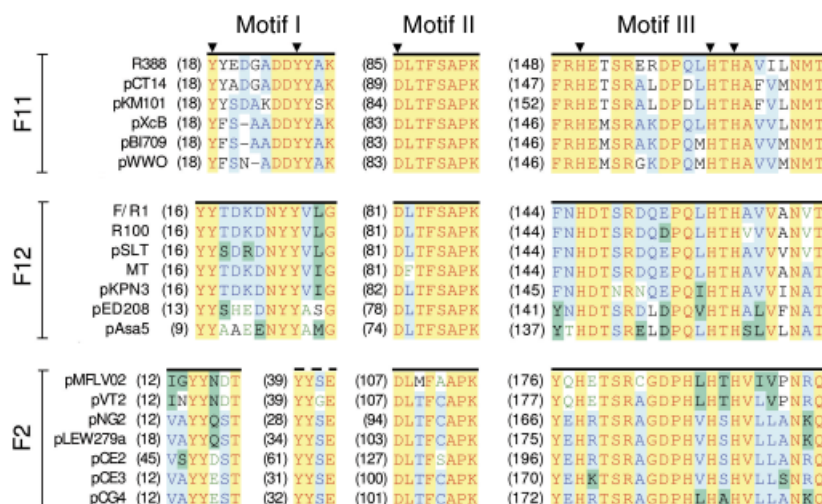


Fig. 4. Conserved sequence motifs in representative relaxase protein clades within the MOB_F family. The clades to which the sequences belong are shown at the left margin of the figure. The CLUSTALW alignment considered only the 300 N-terminal residues of each protein, containing the relaxase domain. Vertical arrowheads point to residues known to be important for function in TrwC_R388 (Guasch *et al.*, 2003). Colour code: red on yellow, invariant amino acids; blue on blue, strongly conserved; black on green, similar; green on white, weakly similar; black on white, not conserved.

The MOB_H family

MOB_H represents an entirely new family, not represented at all among mobilizable plasmids and thus not analysed in our previous review (Francia *et al.*, 2004).

The MOB_H phylogeny tree

The prototype MOB_H relaxase is TraI from the IncHI1 group plasmid R27 (TraI_R27). It was used as a query in PSI-BLAST, resulting in 25 plasmid hits (Table S2) as well as a large number of hits on ICEs and genomic islands (GIs). The search converged in the third iteration. Thus, the MOB_H family is a new and well-resolved relaxase family. It bears no relationship with any other relaxase family. Even today, it is only composed of relaxases from large conjugative plasmids of the classical incompatibility groups (IncH, IncJ, IncT, IncP7 and IncA/C) and genomic elements derived from them. Each of these Inc groups represents a well-resolved clade within the MOB_H tree (Fig. 5). Thus, the phylogenetic relationship among these groups is made clear. Another important hallmark of the MOB_H family is that some clades are composed entirely of ICEs and GIs. Entering the field of chromosomally integrated GIs will add insurmountable complexity to this review; hence, the analysis of conjugative integrated elements had to be confined to a minimum. Because it is likely that a number of GIs may contain nonfunctional relaxases, their inclusion may corrupt the phylogenies of plasmid relaxases, logically assumed to be functional. Nevertheless, a related phylogenetic tree emphasizing the ICEs and GIs in MOB_H has been reported (Salgado-Pabon *et al.*, 2007).

Considering mainly plasmid relaxases and active ICEs, the MOB_H group is resolved into two well-supported clades (MOB_{H1} and MOB_{H2}) bringing together plasmids of

Gammaproteobacteria, and a less resolved and more distant clade MOB_{H3} composed of plasmids of *Betaproteobacteria*. Representatives of clade MOB_{H1} show considerable genetic divergence. In the NJ tree, we can distinguish at least two MOB_{H1} subclades that are not well split in the ML tree (data not shown).

Class MOB_{H11} brings together relaxases belonging to IncHI conjugative plasmids (R27/pHCM1 and R478), peculiar in that conjugation occurs at 30 °C, but not at 37 °C. A second clade (MOB_{H12}) is much broader in phylogenetic terms and is divided in several sub-subclades. One sub-subclade joins together IncA/C plasmids with a group of ICEs [R391 from *Proteus rettgeri* (Coetzee *et al.*, 1972) and SXT from *Vibrio cholerae* (Waldor *et al.*, 1996)]. These two elements combine a set of conjugative transfer genes with a phage-based integration system. Comparison of the R391 and SXT DNA sequences showed conservation of a backbone region, with > 95% identity over 65 kb (Beaber *et al.*, 2002). Both R391 and SXT are capable of excision and subsequent conjugative transfer to recipient cells. R391 was initially classified as an IncJ plasmid, but the absence of any plasmid replicon (Boltner *et al.*, 2002) confirmed that it is not a plasmid. ICEs differ from archetypal conjugative transposons, such as Tn916, in the absence of random chromosomal integration. In fact, R391 and SXT each integrate into a single site in the *Escherichia coli* chromosome. In this respect, they resemble GIs, which also integrate into just one or a few chromosomal sites (typically tRNA genes). Although ICEs are chromosomally located, they have been included in our analysis because they excise to form circular (nonreplicating) plasmids as part of their normal life cycle. The relaxase genes of R391 and SXT were identified, but not analysed experimentally.

Another MOB_{H12} sub-subclade includes plasmids Rts1 and pCAR1. Rts1 (Murata *et al.*, 2002), a 217-kb conjugative

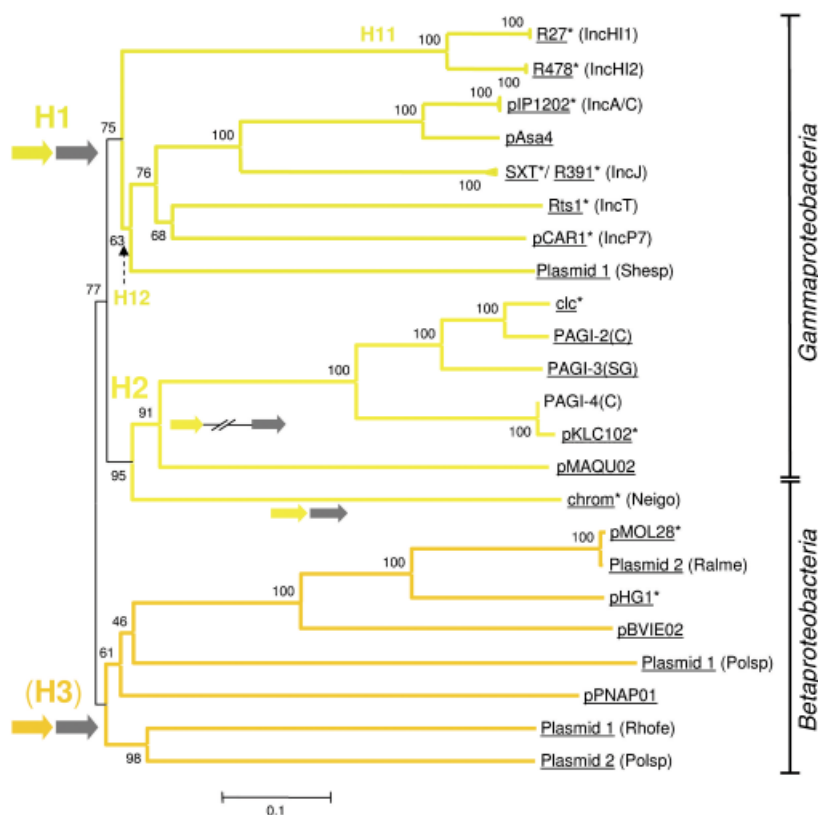


Fig. 5. Phylogeny tree of MOB_H relaxases. The tree is rooted with the MOB_{H3} relaxases. Codes for underlines, asterisks, boxes, gene synteny and vertical bars are the same as in Fig. 2. Clade MOB_{H3} is in parentheses because its status is uncertain. Each clade is coloured in different tones of yellow. Shesp, *Shewanella* sp.; Neigo, *Neisseria gonorrhoeae*; Ralme, *Ralstonia metallidurans*; Polsp, *Polaromonas* sp.; Rhofe, *Rhodoferrax ferrireducens*.

plasmid originally isolated from *Proteus vulgaris*, is the prototype IncT plasmid. Rts1 conjugation is also thermo-sensitive, being most efficient at 25 °C. Conjugation-related T4SS gene products show similarities to F and R27 conjugation systems. Homologues of all F genes for pilus assembly except *trbI* were identified in Rts1. Besides TraI (Orf201), which belongs to MOB_{H12} , there is another protein called Orf220 (accession no. NP_640180), which was annotated as a 'nickase', related to relaxase TaxC_R6K. However, detailed inspection of Orf220 showed that it only contains 7% identity to the C-terminus of TaxC. Plasmid pCAR1 (Maeda *et al.*, 2003) is a 199-kb plasmid isolated from *Pseudomonas resinovorans*. It is the prototype IncP7 plasmid. pCAR1 contains transposon Tn4676 (Shintani *et al.*, 2003), which contains a gene product annotated as 'putative DNA nickase' (Orf 145, accession no. NP_758688). Orf145 is homologous to TraO_pIPO2T, which is actually a DNA primase. These are two examples (among many) of poorly curated annotations that complicate database searches to a great extent.

Besides two relaxases, TraI_pKLC102 and TraI_pMAQU02, belonging to plasmids of *Pseudomonas aeruginosa* and *Marinobacter aquaeolei*, respectively, clade MOB_{H2} is constituted of chromosomally located relaxases. They form part of ICEs, some closely related to plasmid pKLC102, as

shown in the phylogeny of Fig. 5. It is assumed that these ICEs directly originated from ancestral elements closely related to pKLC102 (Klockgether *et al.*, 2004). Interestingly, in *P. aeruginosa* C strains, plasmid pKLC102 was found both as a plasmid and as a chromosomally integrated element. This situation could depict an ongoing evolution from a plasmid to an irreversibly fixed GI, because certain C isolates contained only the chromosomally integrated version of pKLC102 while another subgroup exhibited a diversity of transposition events affecting the integrity of pKLC102 and contributing to its fixation (Klockgether *et al.*, 2004). T4CP genes of MOB_{H2} plasmids and ICEs are located far from the relaxase gene (Fig. 5).

Also included in clade MOB_{H2} is the prototype relaxase TraI of a GI in the *Neisseria gonorrhoeae* chromosome, which will be discussed below. Similar GIs have been detected in several organisms harbouring HD-hydrolases homologous to TraI_R27, such as *Xylella fastidiosa*, *Xanthomonas campestris*, *Burkholderia fungorum* and *Ralstonia metallidurans* (van der Meer & Sentchilo, 2003). These and other GIs and their relaxases were arranged phylogenetically by Salgado-Pabon *et al.* (2007) and will not be described here.

There might be a third clade, MOB_{H3} , that includes the relaxases of conjugative plasmids pHG1 (*Ralstonia eutropha*) and pMOL28 (*R. metallidurans*). These plasmids, as

well as plasmid 2 (*Polaromonas* sp.), plasmid 2 (*R. metallidurans*), plasmid 1 (*Rhodospirillum rubrum*), plasmid 1 (*Polaromonas* sp.) – not included in the same group by ML analysis, pBVIE02 (*Burkholderia vietnamiensis*) and pPNAP01 (*Polaromonas naphthalenivorans*) (all from *Betaproteobacteria*), do exhibit neither the histidine triad nor the HD-hydrolase motif. Nevertheless, they all contain a T4CP adjacent to the relaxase gene and always appeared in BLAST searches starting from plasmids of other MOB_H clades. When a BLAST search was started from the relaxase of pHG1, for instance, it also converged upon retrieving all MOB_H plasmids. Thus, these plasmids are bona fide MOB_H plasmids and we need to consider what this result implies on the structure and mechanism of MOB_H relaxases.

The phylogeny of the MOB_H T4CPs is shown in Fig. S1. As can be seen when comparing Fig. 5 and Fig. S1, MOB_H relaxase and the T4CP phylogenies are congruent, with the exception of MOB_{H3}, which was not well resolved in the relaxase tree either. The branches in the MOB_{H11}, MOB_{H12} or MOB_{H2} clades are completely conserved. This result indicates, once again, that relaxase and T4CP evolve together.

MOB_H relaxase signatures

As explained above, MOB_H relaxases are clearly different from the other relaxase families. A CLUSTALW alignment of MOB_H relaxases that includes clade MOB_{H3} shows no invariant amino acid. Nevertheless, MOB_{H1} and MOB_{H2} clades share two sequence motifs, as shown in Fig. 6: a variant of the 3H motif (that we call alternative 3H motif) and a new, family-specific, HD hydrolase motif. The alternative 3H motif is a strongly conserved sequence of signature (HQ)-x₂-PASE-x-HHH-x₃-GG-x₃-H-x-L, although the sequence environment is very different from that of the 3H motif (compared with MOB_F relaxases in Fig. 3). The HD hydrolase motif contains the signature (LV)-x-HD-(AVLI)-GK. The common activity of proteins showing the HD-motif is that of a divalent cation-dependent phospho-

hydrolase (Aravind & Koonin, 1998). Thus, it is possible that the 3H and the HD motifs are alternative forms of a metal binding cluster. However, until experimental evidence becomes available, the functional status of these motifs remains uncertain. Finally, it was not possible for us to find a motif equivalent to motif I of MOB_F or MOB_P plasmids. Y63 is an invariant Tyr in the MOB_{H1} clade, but this residue is not conserved in clade MOB_{H2}. Curiously, a different Tyr (Y47) was potentially assigned as the catalytic Tyr by Sherburne *et al.* (2000). Y63 neither coincides with the proposed catalytic Tyr of the chromosomal relaxase of *N. gonorrhoeae* (Y93). Although mutants in Y93 are nonfunctional, this residue is not conserved in MOB_{H1} or MOB_{H2} clades (several relaxases contain Phe in this position; see Fig. 6). The MOB_{H3} clade is intractable for signature search, because there is not a single invariant residue. At the present time, the structure and mode of action of MOB_H relaxases are a mystery that deserves biochemical analysis.

MOB_H prototypes

The available experimental information on MOB_H relaxases is scarce. In fact, not a single plasmid conjugative relaxase has been characterized at the biochemical level. The IncHI1 plasmid R27 has been analysed in most detail (Sherburne *et al.*, 2000). This 180-kb-long plasmid is temperature sensitive for conjugation, with optimal mating temperature at 22–28 °C due to transcriptional regulation of its *tra* genes (Alonso *et al.*, 2005). Most genes coding for its T4SS are encoded in one plasmid region called Tra2 (Sherburne *et al.*, 2000), whereas the *oriT*, all relaxosomal components and some additional T4SS genes are encoded in Tra1 (Lawley *et al.*, 2002, 2003). While Tra1 exhibits organizational similarities to both IncP and IncW transfer regions, Tra2 resembles the transfer region of plasmid F. The R27 MOB region was examined by mutagenesis in Lawley *et al.* (2002). The genetic organization is *traR* << *oriT* >> *traH* (nicking accessory protein) >> *traI* (relaxase) >> *traG* (T4CP) >>

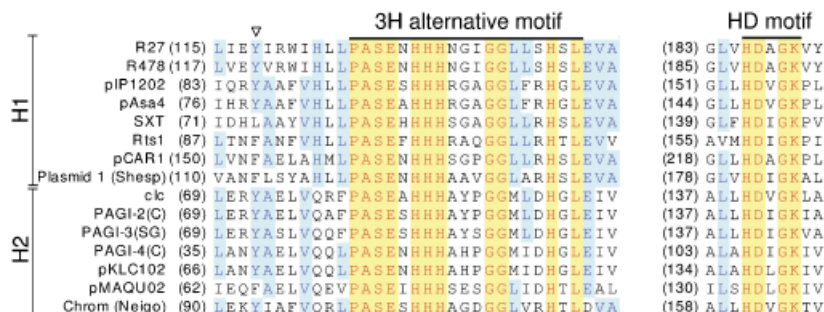


Fig. 6. Conserved sequence motifs in the MOB_H family. Codes are the same as in Fig. 3. The conserved motifs in MOB_H relaxases are different from those in the classical relaxases (like MOB_F). The first motif is histidine-rich; hence, it is named alternative 3H motif to point out that it has no homology to the classical 3H motif. The second motif is shared by HD-hydrolases and it is named accordingly. See MOB_H relaxase signatures for more details. The white vertical arrowhead points to Y93, an important residue of Tra₁Neigo, which is the only MOB_H protein that has been analysed genetically.

traJ (nicking accessory protein). The functional *oriT* is no more than 285 bp long, but no *nic* site was identified. Unfortunately, there are no reports of the biochemical analysis of the predicted relaxase (1011 amino acids). The existence of two potential nicking-accessory proteins also makes the MOB_H system more related to F. Another similarity is the fact that plasmids containing MOB_H and MOB_{F12} relaxases transfer with equal frequencies in liquid and solid surfaces (Bradley *et al.*, 1980).

The most striking system in the family is the DNA secretion system of *N. gonorrhoeae*, included in a 57-kb GI called gonococcal genetic island (GGI) (Hamilton *et al.*, 2005). The GGI island contains a T4SS related to IncF and IncH plasmids that is still mobile. It can be excised and inserted experimentally. The GGI element is responsible for chromosomal DNA secretion to the exterior of the cell. The secreted DNA is then a substrate for the *Neisseria* DNA uptake system. Thus, *Neisseria* seems to have developed a mechanism for DNA release/uptake that allows gonococcus strains to share DNA seamlessly resulting in an effective panmictic clone (Hamilton & Dillard, 2006). Genetic integrity and expression of the T4SS components is required for DNA secretion. The GGI relaxase (TraI_Neigo) and the DNA processing reactions have been studied in detail (Salgado-Pabon *et al.*, 2007). DNA is secreted by the T4SS as ssDNA, as judged by its sensitivity to ss- but not ds-specific DNases. In addition, the DNA is protected in its 5'-end, possibly by the secreted relaxase. The relaxase itself has some interesting and unique properties. Mutation of TraI identified Tyr93 as required for DNA secretion, making it the likely tyrosine that performs the nucleolytic attack on the DNA, by analogy to other relaxases. Residue Tyr201 was proposed as a possible second nucleophile, although results seemed inconclusive. More puzzling was the fact that two residues in the His-triad, also pervasive in conjugative relaxases, could be mutated without affecting the secretion activity (even the double mutation H106A/H108A showed wild-type levels of DNA secretion). Although DNA secretion is a clearly different mechanism from conjugation *per se*, the lack of implication of the 3H constellation in the release mechanism is difficult to conciliate with the current knowledge of the conjugative DNA-processing reactions. The authors pointed out that the conserved HD domain in relaxases of the TraI_Neigo family is known to bind divalent cations (Aravind & Koonin, 1998) so that it could substitute for the 3H domain. Unfortunately, mutations H161S or D162N did not affect DNA release either, casting further doubts on this interpretation. Nevertheless, the D120N mutation clearly affected DNA secretion. This residue was proposed by the authors to be analogous of D85 in protein TrwC (see section The MOB_F family). Finally, TraI_Neigo still has another unique characteristic. Its N-terminal sequence resembles a signal sequence. Although it was demon-

strated that TraI_Neigo is not secreted via the general protein secretion system (Sec-dependent), it is probably bound to the cell membrane by means of its N-terminal peptide. Thus, TraI_Neigo is a membrane-associated protein, contrary to most characterized conjugative relaxases. Lastly, the GGI element *oriT* is tentatively ascribed to an IR close to the relaxase gene (between genes *yaf* and *ltgX*), because an insertion at this point resulted in a non-DNA-secreting phenotype. In summary, the *Neisseria* DNA secretion system constitutes the best analysed of the MOB_H systems. However, not being a truly conjugative system, caution has to be taken before extrapolating results obtained in this system to MOB_H conjugative systems. Conversely, the betaproteobacterial plasmids in MOB_{H3}, which are so poorly characterized in their transfer properties, might be using deviant conjugation mechanisms, perhaps similar to that suggested by the properties of the *Neisseria* DNA release system.

The MOB_C family

The MOB_C family is also a well-separated relaxase family. Two MOB_C relaxases have been analysed experimentally: MobC_CloDF13 (Nunez & de la Cruz, 2001) and TraX_pAD1 (Francia & Clewell, 2002b). A PSI-BLAST search using MobC_CloDF13 as a query and a threshold of $P < 10^{-4}$ converged at iteration 11, retrieving 54 proteins. Twenty-one were from unfinished bacterial genomes and, therefore, are not included in Table S3. The same hit list of 32 proteins (24 corresponding to plasmid relaxases) was obtained when TraX_pAD1 was used as a query. Besides 'canonical' MOB_C relaxases, the hit list included four proteins encoded by small Gram-positive plasmids, which were annotated as RCR initiator proteins (probably because no ORF resembling a Rep protein was found in those plasmids). Nevertheless, the appearance of a downstream T4CP (*mobB*-like) gene in the same plasmids may indicate that they are in fact MOB_C relaxases, as will be discussed later.

The phylogeny tree of the MOB_C relaxases is shown in Fig. 7. Both conjugative and mobilizable plasmids are included, as well as demonstrated ICEs or related elements showing high homology and conserved gene synteny. The family contains elements isolated from *Gammaproteobacteria*, *Firmicutes* and *Tenericutes*. There are three well-resolved clades in the MOB_C phylogeny, according to NJ analysis. Clade MOB_{C1}, of which MobC_CloDF13 is the prototype, is composed of a series of plasmids and other genetic elements from *Gammaproteobacteria*. The case of MOB_{C1} is exceptional in that some of these plasmids contain T4CPs but are mobilizable, not conjugative. This is the unique conspicuous exception to the rule in all plasmids we analysed. Clade MOB_{C2} has conjugative plasmid pAD1 as a prototype and includes relaxases from Gram-positive bacteria (phyla *Firmicutes* and *Tenericutes*). Subclade MOB_{C21} is composed of

enterococcal relaxases while subclade MOB_{C22} includes a number of as yet nonanalysed putative relaxases from *Spiroplasma*. Clade MOB_{C3} – not well resolved by ML analysis – comprises the above-indicated putative RCR replication proteins, which are the most divergent in sequence. T4CP genes are usually found adjacent to and upstream from relaxase genes (Fig. 7). Exceptions included the MOB_{C1} plasmid pVS54, the MOB_{C2} plasmids pSci5, pSci6 and pSci1, in which no T4CP was found, and clade MOB_{C3} plasmids, which contained T4CP genes downstream from the relaxase gene. The *oriT*s, when known, were located upstream of the corresponding *mob* genes, except in the enterococcal members (MOB_{C21}), where the *oriT* was located between oppositely transcribing *mob* genes.

The phylogeny tree of MOB_C T4CPs is shown in Fig. S2. Comparison of this figure with Fig. 7 shows that both trees are highly congruent, showing evident conservation of the basic clades/subclades MOB_{C1}, MOB_{C21}, MOB_{C22} and MOB_{C3}. The only exception is plasmid p4028 whose T4CP clusters closer to the *Spiroplasma* homologues.

Genetic elements CloDF13 (Nunez & de la Cruz, 2001), pAD1 (Francia & Clewell, 2002b), pAM373 (De Boever *et al.*, 2000), pTEF1 (Coburn *et al.*, 2007), ICE EcoR31 (Schubert *et al.*, 2004), ICE Kp1 (Lin *et al.*, 2008) and p29930 (Strauch *et al.*, 2003) code for demonstrated relaxases. Conjugative plasmid pAD1 and the pathogenicity island (PAI) of *Enterococcus faecalis* V583 code for transfer systems with about 95% DNA identity, raising the

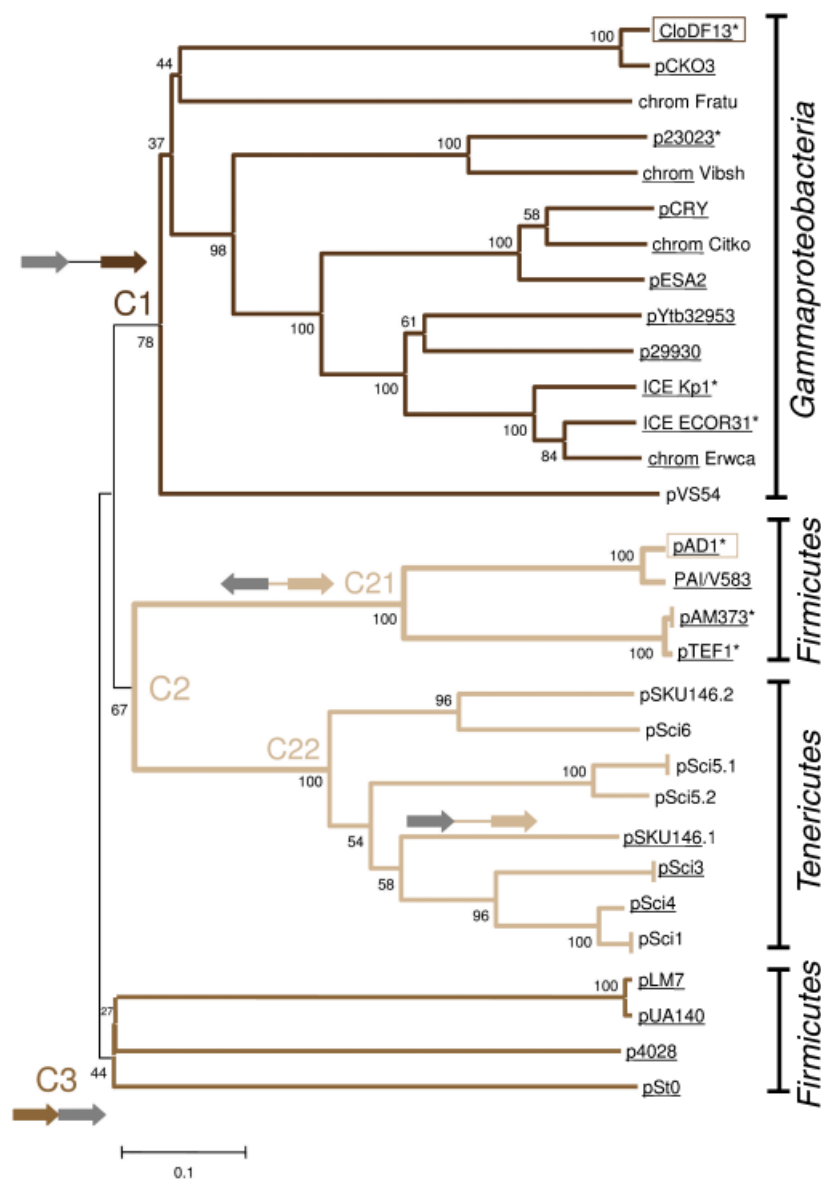


Fig. 7. Phylogeny tree of MOB_C relaxases. Important clades (C1–C3) and subclades (C21 and C22) are highlighted in brown colours. Codes for underlines, asterisks, boxes, gene synteny and vertical bars are the same as in Fig. 2. Fratu, *Francisella tularensis*; Vibsh, *Vibrio shilonii*; Citko, *Citrobacter koseri*; Erwca, *Erwinia carotovora*.

possibility of the enterococcal PAI being self-transmissible (not demonstrated experimentally). The Mob regions in plasmids pTEF1 and pAM373 are closely related and both were shown to be conjugative (De Boever *et al.*, 2000; Coburn *et al.*, 2007). T4CP and relaxase gene homologues were found in the last four enterococcal genetic elements on opposite sides of *oriT* following the same synteny. *Yersinia* cryptic plasmids pCRY and pYtb32953 encoded relaxases and T4CPs homologous to the corresponding genes in plasmids CloDF13 and p29930. Moreover, they encode a T4SS related to that of p29930, suggesting that they might be conjugative. p29930 was shown to be conjugative between *Yersinia* strains and between *E. coli* and several *Yersinia* strains (Strauch *et al.*, 2003). Several MobC homologues are present in GIs of *E. coli*, *Erwinia carotovora*, *Klebsiella pneumoniae*, *Citrobacter koseri* and *Vibrio shilonii*. Self-transmission of the *E. coli* and *Klebsiella* GIs was confirmed experimentally following excision from the chromosome (Schubert *et al.*, 2004; Lin *et al.*, 2008) warranting their classification as ICEs. *Erwinia* GI HAI7 carries all components of relaxosome and T4SS, all highly related to those present in ICE EcoR31 (Evans *et al.*, 2009). Gene synteny is maintained in the mobilization region of most proteobacterial genetic elements, with *oriT* located upstream the *mob* genes. Interestingly, these elements are widespread among *Enterobacteriaceae* members, including a number of *Yersinia enterocolitica* and *Yersinia pseudotuberculosis* clinical isolates. Thus, their contribution to the transmission of virulence factors has been suggested (Schubert *et al.*, 2004; Lin *et al.*, 2008). A growing number of *E. faecalis* and *Enterococcus faecium* clinical isolates contain antibiotic resistant and virulence plasmids with relaxases similar to TraX_pAD1, underscoring the importance of MOB_C relaxases in antibiotic resistance and virulence determinants dissemination (M.V. Francia *et al.*, unpublished data) (Clewell, 2007).

Inspection of the alignment of MOB_{C1} and MOB_{C2} relaxases reveals a conserved signature D-x₆₋₁₇-E-x-E-(RL)-x₂-K-x₃-R-(YF) in a region with strong conservation in adjacent residues (Fig. 8). This is the only invariant signature in the family, further indicating that MOB_C relaxases do not contain equivalents to motifs I and III of the other relaxase families. The signature does not degrade significantly when MOB_{C3} plasmids are included; thus, it could be considered a diagnostic signature for the whole MOB_C family. As a consequence, MOB_C relaxases appear to be unrelated to the relaxases of the other five families.

To provide a further insight into the mechanism of action of MOB_C relaxases, MobC_CloDF13 and TraX_pAD1 were analysed in our laboratories. Site-directed mutagenesis of conserved TraX_pAD1 amino acid residues D152, E170, E172, K176, R180 and Y181 revealed that they were essential for transfer activity (M.V. Francia *et al.*, unpublished data). As a result, Y181_pAD1 (see Fig. 8) is proposed to provide

the nucleophile for the attack of the scissile phosphodiester bond. It has to be noted that *Spiroplasma* relaxases, containing Phe instead of Tyr at the equivalent position, have not been assayed experimentally. The triad composed of residues D152, E170 and E172 in TraX_pAD1 might fulfil a Mg⁺²-coordinating function similar to the His triad (or the HD motif) in other relaxase families.

Another unique characteristic of MOB_C relaxases, demonstrated with MobC_CloDF13, is that the relaxase does not remain covalently bound to the nicked CloDF13 DNA upon cleavage, providing the only reported case of a free (non-protein-bound) nicked DNA molecule as an intermediate in a conjugation or mobilization system (Nunez & de la Cruz, 2001). Moreover, MobC could not relax CloDF13 DNA *in vivo* in the absence of MobB, suggesting a dual participation of MobB in CloDF13 mobilization. Besides its function as a T4CP, it could also be regarded as a nicking-accessory protein for MOB_C activity. It is tempting to speculate a similar helper function for *mobB*-like products in the case of the four putative RCR proteins. In this respect, the *mobB* homologue (*orf2*) of the streptococcal plasmid pUA140 was shown to be essential for stable plasmid maintenance along with *orf1* (*mobC*-like gene) (Zou *et al.*, 2001).

The *oriT* sequences of MOB_C plasmids and ICEs show a large inverted repeat adjacent to a series of short direct repeats. Although some plasmids show conservation of the stem-loop structure, where the *nic* site was located for the prototype plasmids CloDF13 and pAD1, specificity in the nicking reaction was proposed to be determined by the interaction of the respective relaxases with the direct repeats (Francia & Clewell, 2002b).

The MOB_Q family

A PSI-BLAST search (threshold $e < 10^{-4}$), which queried the microbial sequence databases with the sequence of MobA_RS1010 relaxase, the known MOB_Q prototype (Francia *et al.*, 2004), converged in the fifth iteration. One hundred and fifteen hits were retrieved, originating from mobilizable as well as from conjugative plasmids (Table S4). From the first iteration, the family included a number of relaxases that also appeared in the BLAST search querying TraI_RP4 (the MOB_P prototype), indicating that MOB_Q and MOB_P overlap. A phylogenetic tree was constructed from the alignment of all MOB_Q sequences, and is shown in Fig. 9. Prototype relaxases of the MOB_Q family are TraA_pTi (the conjugative relaxase of pTi plasmids of *Agrobacterium*, which has to be distinguished from VirD2 relaxase, responsible for T-DNA transfer to plant cells) and TraA_p42a (same from *Rhizobium* plasmids) besides MobA_RS1010 and TraA_pIP501.

Several clades are well resolved in Fig. 9, using both NJ and ML approaches. The first clade, MOB_{Q1}, brings together

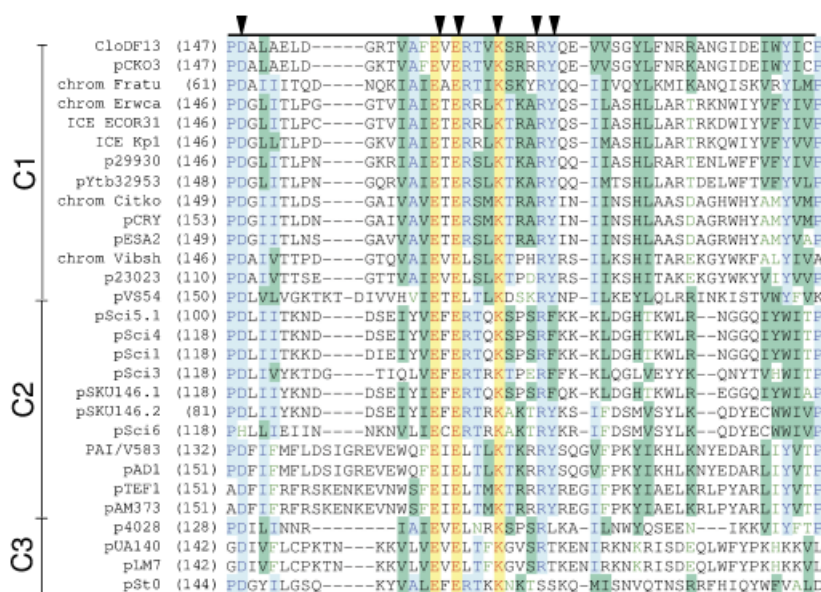


Fig. 8. Conserved sequence motifs in the MOB_C family. Codes are the same as in Fig. 3. pAD1 relaxase essential amino acids (see section The MOB_C family) are represented by black arrowheads.

mobilizable plasmids related to the IncQ plasmid RSF1010. The prototype is MobA_RS1010. The members of this clade are related to MOB_P, because they appeared in a BLAST using TraI_RP4 as a query. The intersection with MOB_P only applied to the MOB_{Q1} clade. The rest of MOB_Q relaxases have further evolved so that similarity with MOB_P was no longer detected by PSI-BLAST when using the standard threshold. MOB_{Q1} relaxases are highly divergent; hence, some internal nodes of this subclade change in the ML topology.

MOB_{Q2} is composed of conjugative plasmids of *Rhizobium* and *Agrobacterium*, among others in the order *Rhizobiales*. The prototype for this clade is relaxase TraA_p42a (Perez-Mendoza *et al.*, 2006). MOB_{Q2} relaxases contain a DNA-helicase domain in their C-terminus, such as MOB_F relaxases. It has to be pointed out that pTi plasmids contain two relaxases. One, called VirD2, is responsible for T-DNA transfer to plant cells and belongs to the MOB_P type. The second, TraA, is responsible for plasmid transfer between *Agrobacterium*, and is the one represented in this group. Another perhaps surprising characteristic of these relaxases is that they seem to act preferentially in *cis* (Perez-Mendoza *et al.*, 2006; Cho & Winans, 2007).

Other clades are also apparent. We will only mention clade MOB_{Q3}, which contains plasmids of phylum *Firmicutes*, the prototype of which is TraA_pIP501 (Inc18). It is the relaxase itself, when binding to the *oriT*, that represses expression of the *tra* operon (Kurenbach *et al.*, 2006). Although pIP501 is found mainly in Gram-positive bacteria, its T4SS is very similar to those of the *Proteobacteria* and apparently contains the same components that assemble a channel traversing both Gram-negative bacterial membranes (Abajy *et al.*, 2007).

The phylogeny tree of MOB_Q T4CPs is shown in Fig. S3. Comparison of this figure with Fig. 9 shows that both trees are highly congruent, showing complete conservation of all branches in the basic clades MOB_{Q2} and MOB_{Q3} (MOB_{Q1} is composed exclusively of mobilizable plasmids, which do not contain T4CPs).

The MOB_Q family was already analysed in detail in our previous review (Francia *et al.*, 2004). Since then, the resolution of the crystal structure of an RSF1010 relaxase fragment (minMobA), in fact the identical protein from the closely related plasmid R1162 (Monzingo *et al.*, 2007), has represented an important advance in our understanding of plasmid relaxases. This is the first relaxase structure that does not belong to MOB_F. The sequences of minMobA_R1162 and TrwC_R388 are < 15% identical. Nevertheless, their three-dimensional structures are very similar. This is an important finding because it links the MOB_F family to the MOB_P cluster. In fact, when looking at the structures, it becomes clear that they contain a homologous catalytic centre, with the His-triad coordinating a divalent metal ion and the catalytic Tyr (Y25 in minMobA_R1162) in its proximity. Besides, MOB_F and MOB_Q sequences share the three classical relaxase motifs, as shown in Fig. 10. There are no arguments about motif I (containing the catalytic Tyr) and motif III (the His-triad), but there is an issue concerning motif II. In the TrwC_R388 structure, residue D85 (which is essential for relaxase activity) is located near the catalytic Y18, possibly abstracting a proton from the hydroxyl group that attacks the phosphodiester bond. In the MobA_R1162 structure, E74 occupies the same position as D85 in TrwC_R388. However, Monzingo *et al.* (2007) have shown that mutation of E74 does not affect mobilization of R1162; hence, they believe that this

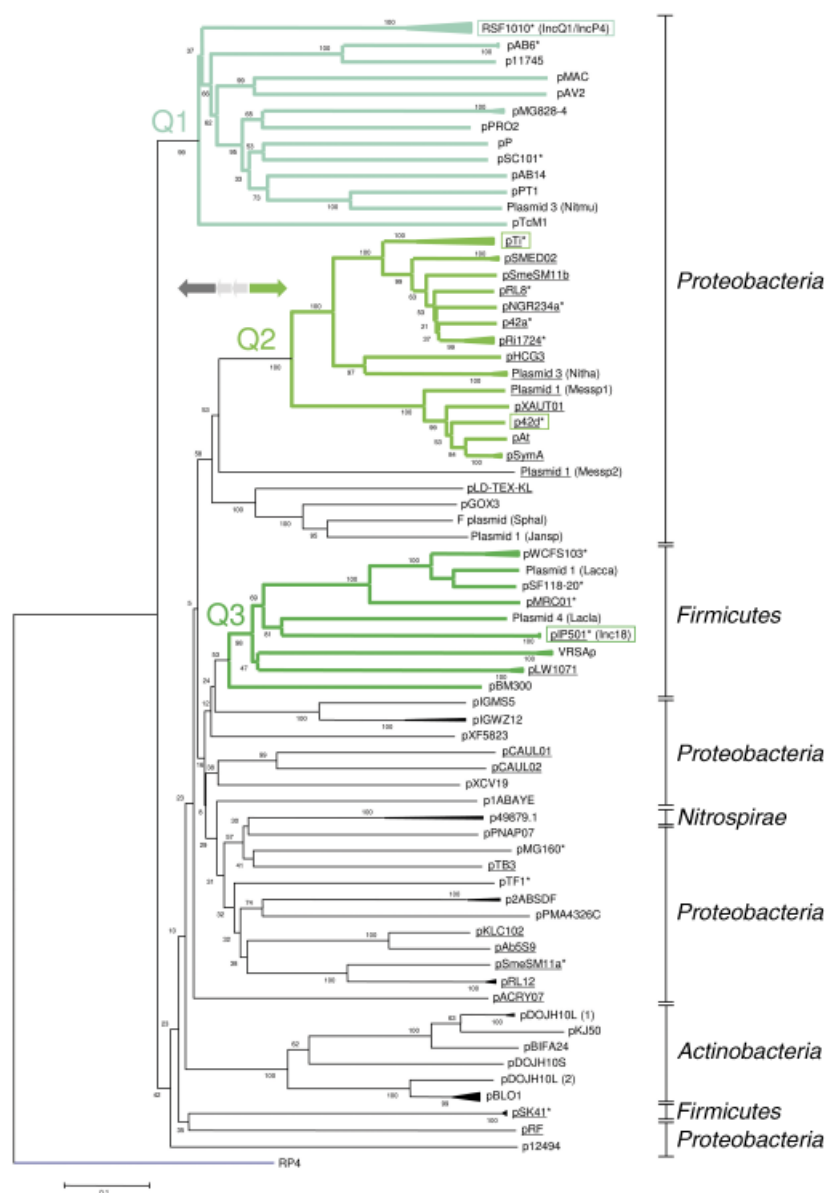


Fig. 9. Phylogeny tree of MOB_Q relaxases. Codes for underlines, asterisks, boxes, gene synteny and vertical bars are the same as in Fig. 2. Nitmu, *Nitrosospira multiformis*; Nitha, *Nitrobacter hamburguensis*; Messp1, *Mesorhizobium* sp. BCN1 – accession number YP_665951; Messp2, *Mesorhizobium* sp. BCN1 – accession number YP_665867; Sphal, *Sphingopyxis alaskensis*; Jansp, *Jannaschia* sp.; Lacca, *Lactobacillus casei*; Lacca, *Lactococcus lactis* [pDOJH10L (1), accession number NP_694600; pDOJH10L (2), accession number NP_694596].

residue is not important. In apparent accordance with these data, E74 is not an invariant residue in Fig. 10. It can be substituted by aspartic acid (a conservative change), but also by serine (in the case of plasmid pAV2). A much more conserved Asp in motif II is E82. Although it lies far from the catalytic centre, E82 is in a flexible loop in the resolved protein fragment and we do not know what its position will be in the complete protein. Unfortunately, its involvement in MobA activity has not yet been tested by mutagenesis. Monzingo *et al.* (2007) propose that the function of TrwC residue D85 is possibly not required in MOB_Q relaxases, because *nic*-cleavage is not the limiting step in plasmid R1162 conjugation. If they are right, this can be an important mechanistic difference between MOB_Q and MOB_F relaxases.

The MOB_P cluster

A PSI-BLAST search using TraI_{RP4} as a query and a score threshold $P < 10^{-4}$ converged after 14 iterations and retrieved 259 relaxase sequences (Tables S4–S6). MOB_P represents therefore the largest relaxase family (see Fig. 1). Among the hits were relaxases belonging to clade MOB_{Q1}, which were discussed in the previous section, plus all the previously described MOB_{HEN} relaxases (Francia *et al.*, 2004), of which MbeA_{ColE1} is the prototype. Thus MOB_P and MOB_Q are neighbouring protein families, while MOB_{HEN} is totally embedded in MOB_P. This notion is represented in Fig. 1 by including MOB_{HEN} within MOB_P and intersecting it with MOB_Q and MOB_V. Thus, MOB_P is a

cluster of actively evolving relaxases, with several distinguishable clades. Figure 11 shows a phylogenetic tree of the MOB_P cluster (collapsed at the branches) that emphasizes its large diversity and pinpoints the prototypical clades: MOB_{P1} (that groups the IncP complex, the IncI complex and the IncQ2/IncG/IncP6 group), MOB_{P2} (the VirD2 proteins from *Agrobacterium* plasmids), MOB_{P3} (the IncX complex), MOB_{P4} (the IncU plasmids), MOB_{P5} (that totally includes the MOB_{HEN} family), MOB_{P6} (the IncI2 plasmids) and MOB_{Q1}, among several other clades whose phylogeny cannot be well resolved yet.

The phylogeny of MOB_{P1} relaxases is shown in more detail in Fig. 12. Its four well-resolved clades are shown by different tones of blue colour in the figure. These subclades are also distinguishable by ML analysis. Clade MOB_{P1} hosts are distributed in all divisions within the *Proteobacteria*. Subclades of relaxases extend also all over the *Proteobacteria*; thus, in this case assigning hosts in the phylogeny of Fig. 12 will not be informative. This promiscuity is probably an important property of MOB_{P1} plasmids and is certainly a characteristic of its most famous representative, plasmid RP4.

Clade MOB_{P11} is composed exclusively of conjugative plasmids and has RP4 (IncP1 α) and R751 (IncP1 β) as prototypes. In fact, the IncP cluster (Heuer et al., 2004; Haines et al., 2006b; Bahl et al., 2007) is broad, and includes probably all plasmids listed in Fig. 12 from RP4 down to 'plasmid 59 kb'. Thus, the IncP group is a deep, well-populated clade that contains a large number of conjugative

plasmids. We propose to call it MOB_{P111}. Because this is probably the most thoroughly analysed plasmid incompatibility group, it serves to underline the spread we can expect of an Inc group when it is analysed exhaustively. Although many other Inc groups are only represented by a single member in this work, each Inc group might be expected to encompass a clade that can reach the extension of IncP or IncF groups.

Clade MOB_{P12} is smaller in depth and breadth than MOB_{P11} and is interesting because it represents the conjugative plasmids of the IncI plasmid complex, being the prototype IncI1 plasmid R64. The IncI complex is formed by IncI1 (=IncI α), IncB, IncK and IncZ (see, for instance, Praszkie et al., 1991; Chilley & Wilkins, 1995). Most relaxases of plasmids belonging to the IncI-complex are very similar to NikB_R64 and are grouped with it in the tree. The exceptions are plasmids R387 and pSERB1 from *Shigella flexneri* and enteroaggregative *E. coli*, respectively. Although plasmid pSERB1 was ascribed to the IncI complex (Dudley et al., 2006), its replication protein (RepA) show 100% identity to that of the IncK plasmid R387 (our data), the only representative of incompatibility group IncK. Thus we can assume pSERB1 is IncK instead of IncI1. Accordingly, NikB_pSERB1 is only 75% identical to NikB_R64, while it is 93% identical to NikB_R387. Thus, in the MOB_{P12} phylogeny, the IncI1 and IncK exemplars (R64 and R387) seem to represent two valid phylogenetic lineages and may represent alternative evolutionary strategies. IncI and IncK plasmids are important contributors to the dissemination of

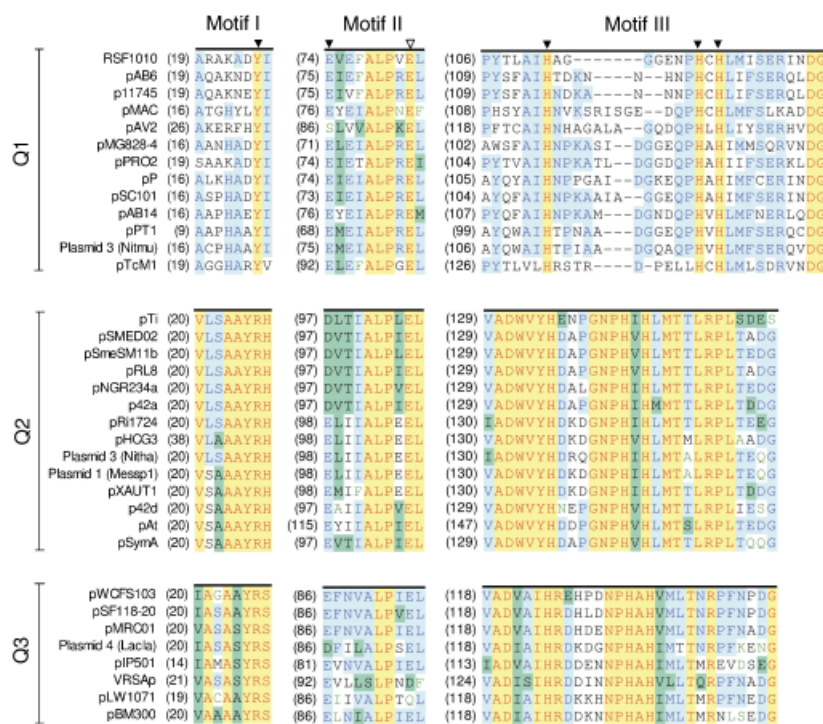


Fig. 10. Conserved sequence motifs in the MOB_Q family. Codes are the same as in Fig. 3. Vertical black arrowheads point to residues configuring the catalytic centre. The vertical white arrowhead points to E82, another potential key residue (see The MOB_Q family).

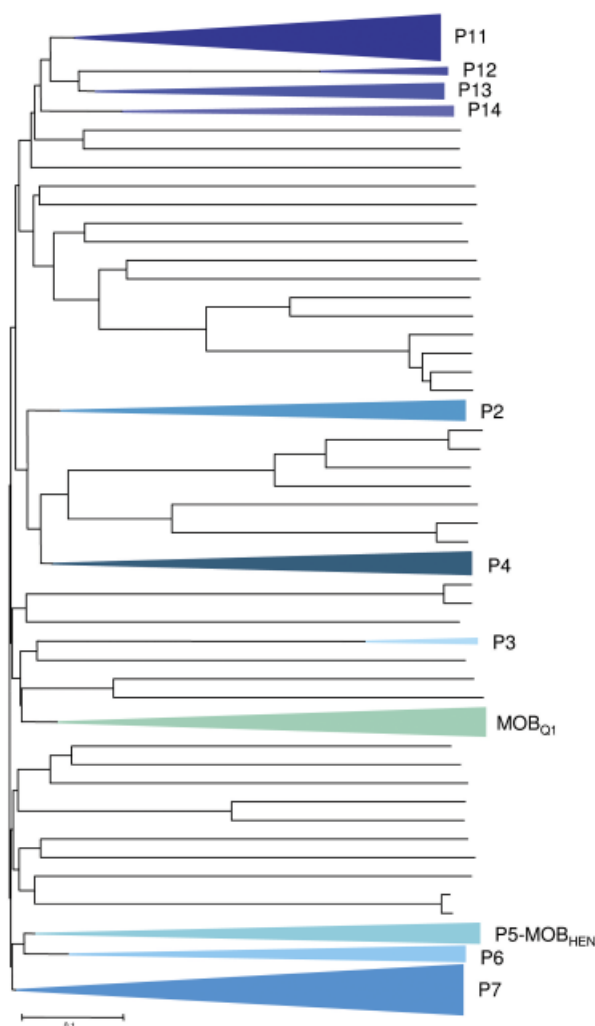


Fig. 11. Phylogeny tree of MOB_P relaxases. In this tree, which contains no names, the terminal branches are collapsed to emphasize the diversity of this relaxase family. The named individual clades are represented in detail in subsequent figures and are represented in different tones of blue.

resistance to extended-spectrum β -lactamases (Clockaert *et al.*, 2007; Navarro *et al.*, 2007), among other antibiotics.

Clade MOB_{P13} contains, among other groups, the IncL/M plasmids. The prototype is the IncL/M plasmid pCTX-M3, which is involved, as many other IncL/M plasmids, in the dissemination of resistance to extended-spectrum β -lactamase genes (Villa *et al.*, 2000; Golebiewski *et al.*, 2007; Novais *et al.*, 2007). The overall homology of MOB_{P13} relaxases is small and some bootstrap values are not high. Thus, it is not surprising that some branches change their position in the ML tree.

Finally, clade MOB_{P14} is composed of a series of mobilizable plasmids; among them, those belonging to the so-called

IncQ2 and IncG/IncP6 incompatibility groups. The IncQ2 plasmids pTF-FC2 and pTC-F14 (Rawlings, 2005) are in fact compatible, although closely related in molecular terms. On the other hand, pTC-F14 and the IncQ1 prototype plasmid RSF1010 are incompatible although less evolutionarily related. This is just one among many examples of the traps that lead to incompatibility analysis. Rms149, the prototype IncG/IncP6 group, replicates both in *E. coli* and *P. aeruginosa*, and thus has a double Inc attribution (Haines *et al.*, 2005). Later, the same research group suggested that IncG should be merged with IncU due to homology among replicons and weak incompatibility between them (Haines *et al.*, 2006a). However, IncU relaxases (like VirD2_pFBAOT6) belong to a different clade (MOB_{P4}). In fact, MobA_Rms149 is about 70% identical to those of IncQ2 plasmids while it is < 25% identical to VirD2_pFBAOT6. Besides, Rms149 replication protein is only 71% identical to that of pFBAOT6. Thus, we suggest that IncG and IncU are preserved, because they denote two differentiated molecular species.

The conserved motifs in MOB_{P1} are shown in Fig. 13. As can be seen, MOB_{P1} relaxases share the classical motifs of MOB_F and MOB_Q relaxases. The most conspicuous is the 3H motif, which shows the signature H-x-(DE)-T-(DE)-x2-H-x-H-x3-N-x3-P. The argument about the acidic residue in motif II, which was invoked in section The MOB_Q family, continues here because MOB_{P1} relaxases also contain a conserved D/E (E80 in TraI_RP4, signalled by a white arrowhead in Fig. 13) that could play the same role as D85 in TrwC_R388. Site-directed mutagenesis of TraI_RP4 (Pansegrau *et al.*, 1994) defined the three sequence motifs in MOB_P relaxases and identified one essential residue in each motif (Y22, S74 and H116). E80 was not tested and, surprisingly, H109 or H118 showed to be nonessential (Fig. 13).

The phylogeny tree of MOB_{P1} T4CPs is shown in Fig. S4. Comparison of this figure with Fig. 12 shows that both trees are basically congruent, showing conservation of the basic clades and subclades. Thanks to the wide knowledge that has been accumulated about this clade, and more specifically the MOB_{P111}-IncP subclade, it will be possible to find the causes for the small discrepancies in some of the terminal branches between both trees.

The phylogeny of the MOB_{P2}, MOB_{P3} and MOB_{P4} relaxases is shown in Fig. 14. NJ and ML topologies are the same with the exception of some deep branches of subclade MOB_{P4}. Among MOB_{P2} relaxases, all of them lying in plasmids from *Alphaproteobacteria*, the most extensively analysed to date are the VirD2 proteins of *Agrobacterium* plasmids. The proteins in *Agrobacterium* play similar roles as relaxases but for transfer of the T-DNA to plant cells in a phenomenon similar to conjugation (Zambryski *et al.*, 1989). The molecular analysis of VirD2_pTiC58 demonstrated that VirD2 proteins are similar to conjugative relaxases in the molecular details of their activity (Pansegrau

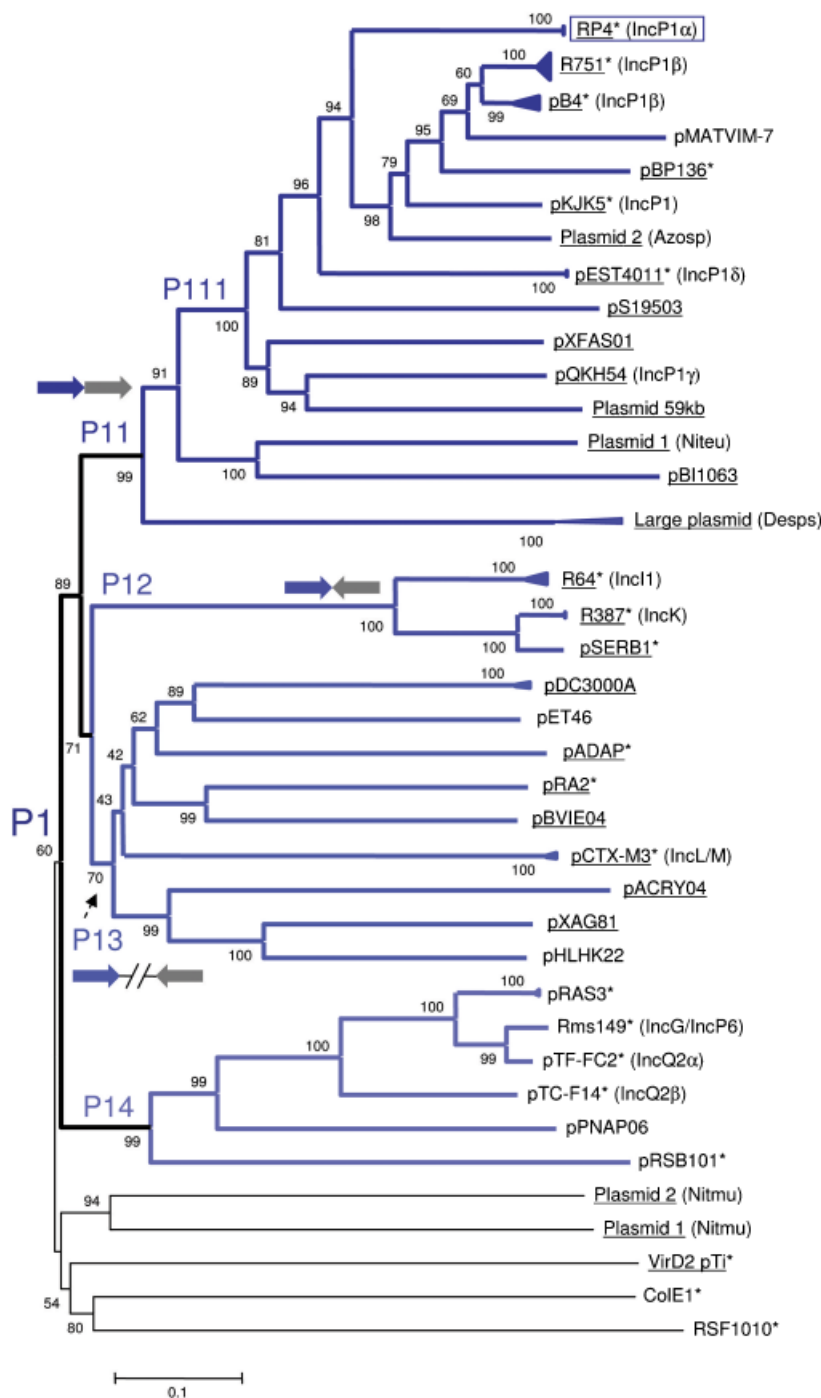


Fig. 12. Phylogeny tree of MOB_{P1} relaxases. Codes for underlines, asterisks, boxes, gene synteny and vertical bars are the same as in Fig. 2. Azosp, *Azoarcus* sp. EbN1; Niteu, *Nitrosomonas eutropha*; Desps = *Desulfotalea psychrophila*; Nitmu = *Nitrosospira multiformis*.

et al., 1993). Besides this, no MOB_{P2} relaxase has been studied in detail.

The MOB_{P3} clade is composed of a series of relaxases from IncX plasmids of *Gammaproteobacteria*, as shown in Fig. 14. TaxC, the relaxase of the IncX2 plasmid R6K (Nunez *et al.*, 1997) can be considered the prototype of the group. Interestingly, plasmid R6K is peculiar in that it contains two

functional *oriTs* (Avila *et al.*, 1996). IncX plasmids were separated in two groups because of lack of hybridization in their replication regions (Jones *et al.*, 1993). However, analysis of their relaxases, as shown in Fig. 14, demonstrates that the IncX group can be considered a real clade, with two branches represented by IncX1 and IncX2 plasmids. The IncX1 plasmids are widespread among opportunistic

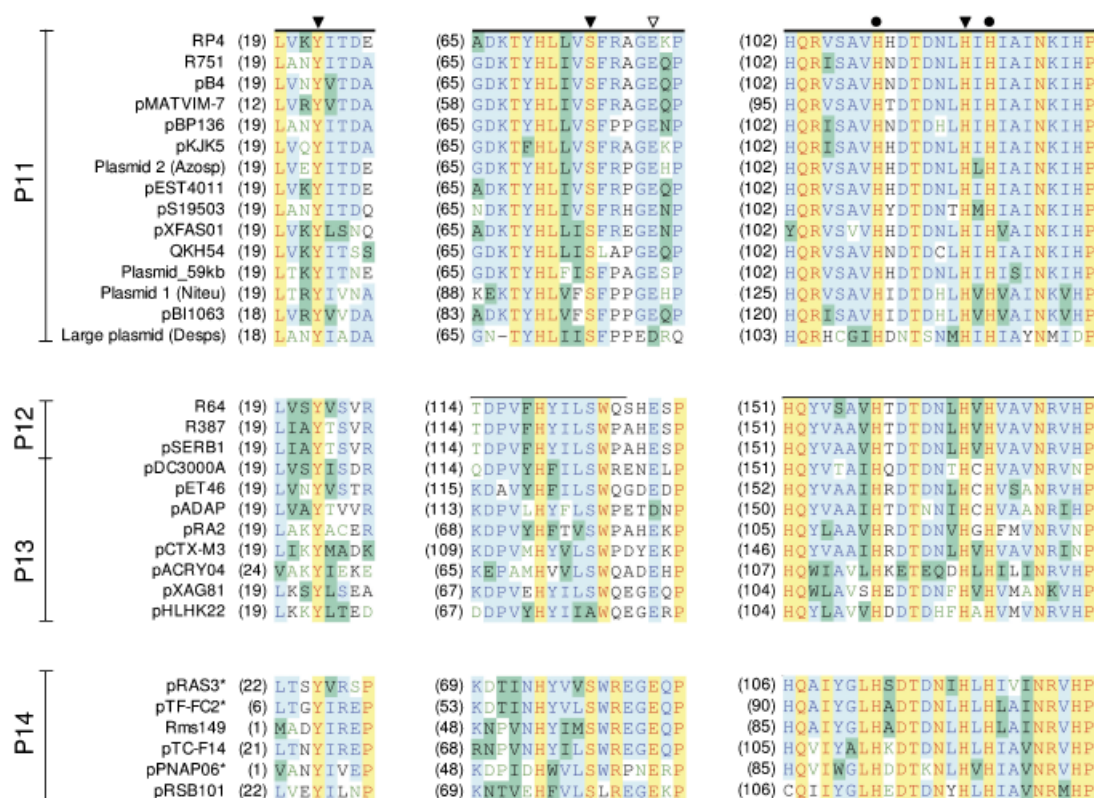


Fig. 13. Conserved sequence motifs in the MOB_{P1} subclades. Vertical black arrowheads and black circles point to residues configuring the catalytic centre. Mutations in TraI_{RP4} affecting those residues either affected conjugation (black arrowheads) or did not (black circles). The vertical white arrowhead points to E79 in TraI_{RP4}, another potential key residue (see The MOB_P cluster). *The N-terminal sequences of these relaxases were wrongly annotated (by comparison with homologous relaxases). Visual inspection allowed us to realize that if the annotated CDSs were extended upstream (pRAS3: – 57 codons; pTF-FC2: – 41 codons; pPNAP06: – 13 codons), motif I was found. Azosp, *Azoarcus* sp. EbN1; Niteu, *Nitrosomonas eutropha* C91; Desps, *Desulfotalea psychrophila* L5v54.

bacterial pathogens, and they code for multiple-drug-resistant efflux pumps as well as the formation of fimbriae (Norman *et al.*, 2008).

The MOB_{P4} clade is extensive, with many deep branches encompassing the various classes of *Proteobacteria*. It includes the IncU plasmids RA3 and pFBAOT6, that are closely related and have a broad host range, being maintained in *Alpha*-, *Beta*- and *Gammaproteobacteria* (Rhodes *et al.*, 2004; Kulinska *et al.*, 2008). Recently, Qnr quinolone resistance genes were found located in IncU plasmids in *Aeromonas* isolates, underscoring the importance of these plasmids in Qnr diffusion outside *Enterobacteriaceae* (Cattor *et al.*, 2008). Other subclade is composed of plasmids isolated from unknown bacteria in soil, the representatives of which are pIPO2 (Tauch *et al.*, 2002) and pSB102 (Schneiker *et al.*, 2001). Other clades contain conjugative plasmids from *Campylobacter* (pTET and pCC31) or *Actinobacillus actinomycetemcomitans* (pVT745), a periodontal pathogen, among others.

The phylogeny of the MOB_{P5}, MOB_{P6} and MOB_{P7} relaxases is shown in Fig. 15. As can be observed in the

phylogenetic tree, the MOB_{P5} clade groups all MOB_{HEN} plasmids together with plasmids pAsa1 and pAsa3, with a bootstrap value of 98%. The MOB_{HEN} subclade contains only mobilizable plasmids and was discussed extensively in our previous review (Francia *et al.*, 2004). Interestingly, the relaxases of plasmids pAsa1 and pAsa3 (together with their close relatives pAsa11, pAsa12 and pPNAP08, not shown in the figure) still contain a classical 3H motif. Thus, we assume the HEN motif arose by mutation of the 3H motif in ancestral plasmids similar to pAsa1 or pAsa3. Once the MOB_{HEN} subclade arose, it was evolutionarily very successful and gave rise to the wide radiation that can be observed in the tree. It will be interesting to determine the functional consequences of this change in the relaxase active site.

Clade MOB_{P6} contains plasmid R721, the prototype of the IncI2 incompatibility group. Not much is known about this plasmid. It produces two types of pili, as IncI1 plasmids: a thin type, not involved in conjugation but in the stabilization of aggregates and a thick type, which allows conjugation in solid surfaces (Bradley, 1984). Besides, R721 contains a shufflon structure, similar to that of IncI1 plasmids that

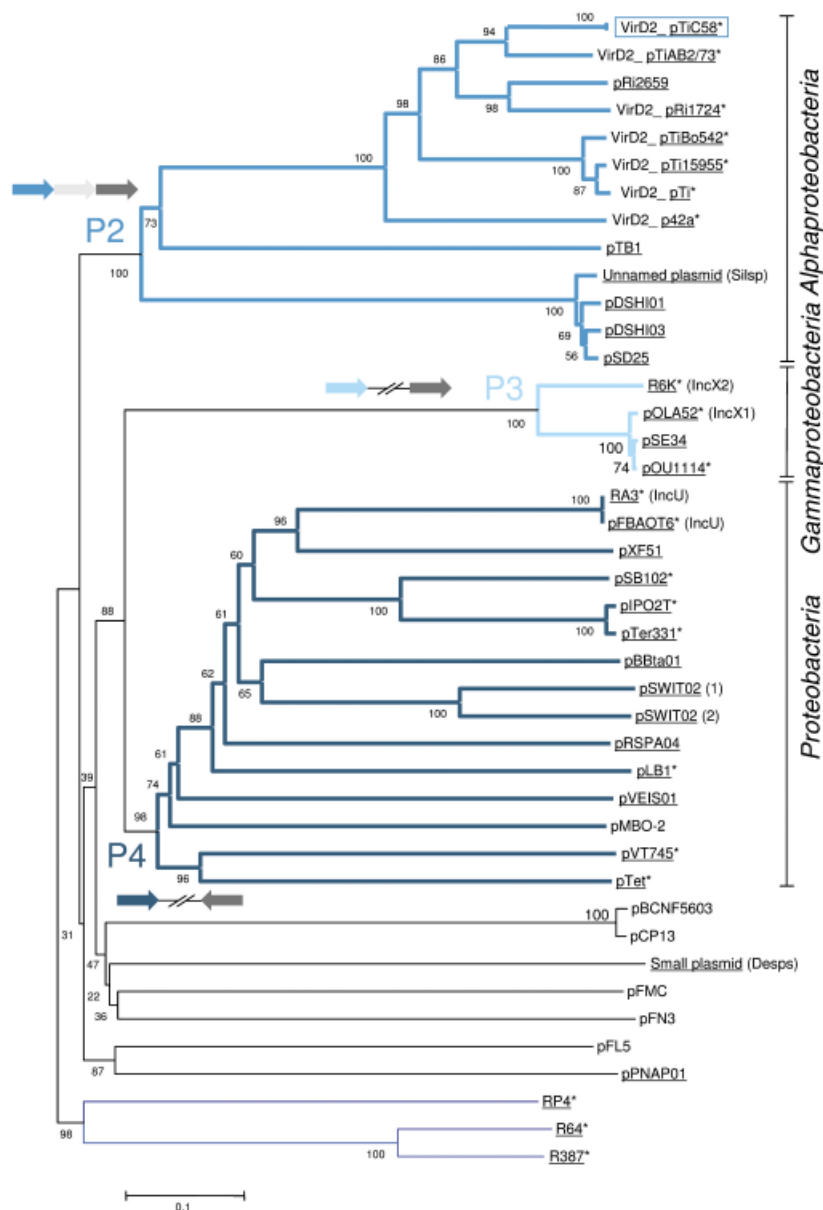


Fig. 14. Phylogeny tree of MOB_{P2+P3+P4} relaxases. Codes for underlines, asterisks, boxes, gene synteny and vertical bars are the same as in Fig. 2. Silsp, *Silicibacter* sp. TM1040; pSWIT02 (1), accession number YP_001260320; pSWIT02 (2), accession number YP_001260361; Dessp, *Desulfotalea psychrophila*.

allows them to alternate some surface antigenic determinants. The shufflon is a DNA region that undergoes complex rearrangement mediated by a plasmid-encoded site-specific recombinase called Rci (Kim & Komano, 1992).

Clade MOB_{P7} is a large clade with profound branches (their resolution is different in NJ and ML topologies). It is the only MOB_P clade, out of all those described above, which extends over several bacterial phyla, including *Proteobacteria*, *Firmicutes* and *Fusobacteria*. Little is known about most of these plasmids. The exceptions are the *E. faecalis* pheromone-responsive plasmid pCF10 (Chen *et al.*, 2007) and two small plasmids from *Staphylococcus aureus* called pC221 and pC223, which can be mobilized by the conjuga-

tive plasmid pGO1 (Caryl *et al.*, 2004; Caryl & Thomas, 2006).

Although not included in any of the above-mentioned MOB_P clades, pEF1 and pHT β -like plasmids deserve a specific comment due to their recognized role in antibiotic resistance dissemination. pHT β -like plasmids are highly conjugative plasmids frequently found in glycopeptide and aminoglycoside-resistant *Enterococcus* clinical isolates in Japan and United States (Tomita *et al.*, 2002; Tomita *et al.*, 2003), while vancomycin resistance was frequently associated with conjugative pEF1-related plasmids in *E. faecium* isolates from different continents (Freitas *et al.*, 2008; Romo *et al.*, 2008).

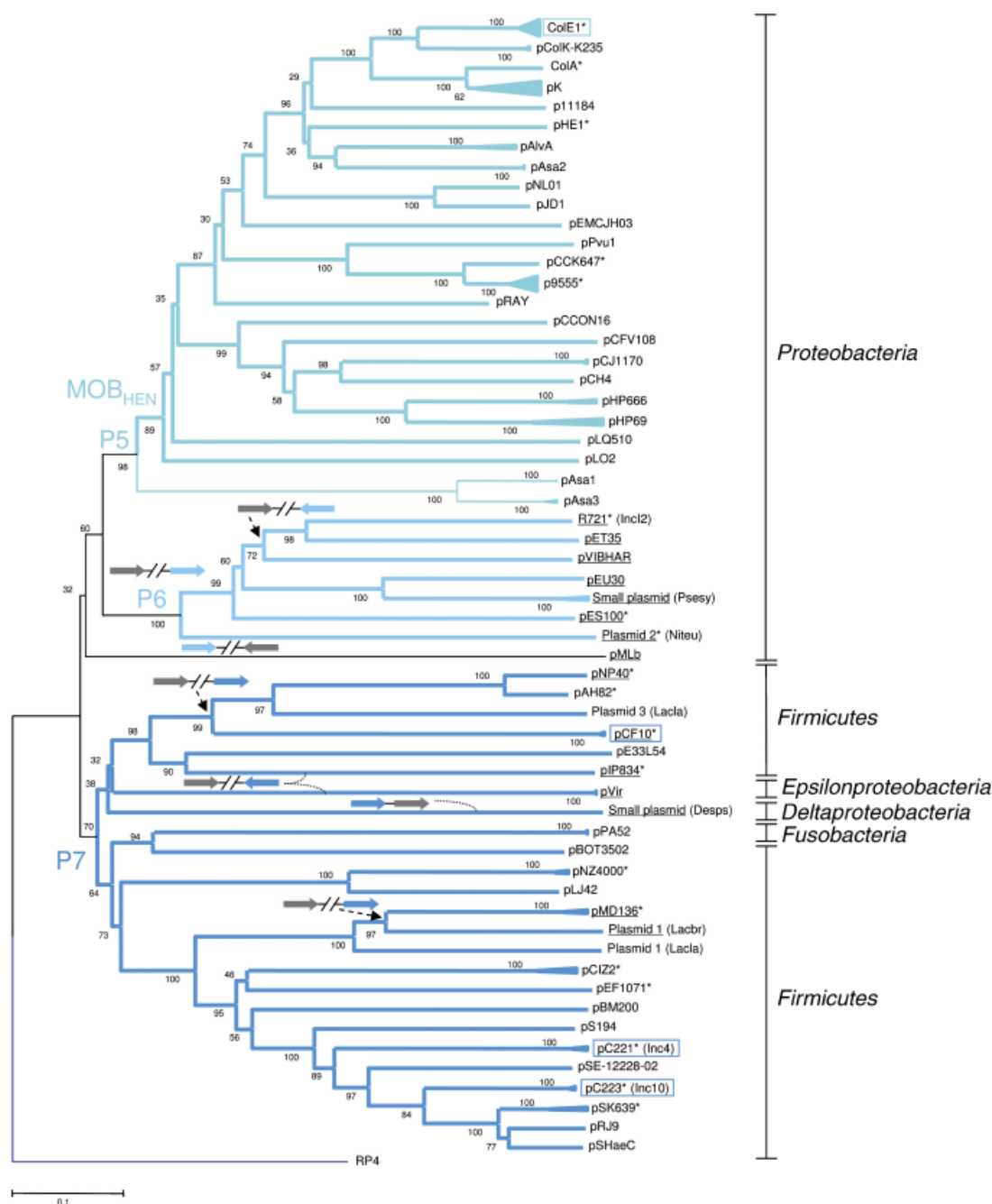


Fig. 15. Phylogeny tree of MOB_{P5+P6+P7} relaxases. Codes for underlines, asterisks, boxes, gene syntenies and vertical bars are the same as in Fig. 2. Psesy, *Pseudomonas syringae*; Niteu, *Nitrosomonas eutropha*; LacIa, *Lactococcus lactis*; Desps, *Desulfotalea psychrophila*; Lacbr, *Lactobacillus brevis*.

Several relaxases of the MOB_P cluster have been analysed biochemically. In general, they are large proteins and contain the relaxase activity in their N-terminal domain. The prototype is the MOB_{P1} relaxase TraI_{RP4} (Pansegrau & Lanka, 1996). Some details of its sequence and properties have been discussed above, when dealing with MOB_{P1} relaxases. Other relaxases that have been analysed are the

MOB_{P1} relaxase NikB_{R64} (Furuya & Komano, 2003), MOB_{P2} VirD2_{pTiC58} (Pansegrau *et al.*, 1993), MOB_{P3} TaxC_{R6K} (Nunez *et al.*, 1997), MOB_{P7} PcfG_{pCF10} (Chen *et al.*, 2007) and MOB_{P7} MobA_{pC221} (Caryl & Thomas, 2006). In general terms, they all behave similarly to TraI_{RP4}. With respect to the analysed MOB_F relaxases, in the MOB_P relaxases there is only one active tyrosine in their

catalytic centre, and this seems to be an important mechanistic difference between MOB_F relaxases and the MOB_P cluster. MOB_P relaxases will probably be structurally similar to MobA_RSFI010, which has been already been discussed (see section The MOB_Q family).

The MOB_V cluster

PSI-BLAST searches were carried out using MobM_pMV158 as a query. When the standard threshold ($e < 10^{-4}$) was used, the set of retrieved proteins included MOB_{P3} and MOB_{P4} relaxases. This result suggests that MOB_V and MOB_P proteins are ancestrally related and that their similarity can still be detected by PSI-BLAST. Iterative BLAST under these settings converged at the 18 iteration, but ended up retrieving many unrelated sequences. When a more stringent $P < 10^{-5}$ threshold was used, the search converged in the eighth iteration, retrieving 105 proteins (Table S7). Ninety-eight hits were plasmid associated while six relaxases were encoded by demonstrated ICEs or IMEs. The MOB_V phylogenetic tree, rooted with TraI_RP4 and a number of MOB_{P3} and MOB_{P4} relaxases, is shown in Fig. 16. MOB_V plasmids have extended along several bacterial phyla, mainly *Firmicutes* and *Bacteroidetes* (where they are known to be associated to various antibiotic resistance genes), but there are also small clades from *Proteobacteria*, *Cyanobacteria* and *Spirochaetes*. Irrespective of their huge diversity, MOB_V elements, including the *Bacteroidetes* transposons, share the same genetic organization of the Mob region and similarities in their *oriT*s (Francia *et al.*, 2004).

MOB_V genetic elements with an experimentally demonstrated relaxase are the *Streptococcus* mobilizable plasmid pMV158 (Priebe & Lacks, 1989; Guzman & Espinosa, 1997) (de Antonio *et al.*, 2004), the *Bordetella* plasmid pBBR1 (Szpirer *et al.*, 2001) and the *Bacteroides* transposons Tn4555 (Smith & Parker, 1998) and Tn5520 (Vedantam *et al.*, 2006). Besides, there is experimental evidence of mobilization for additional 21 plasmids in *Firmicutes*, *Bacteroidetes* and *Proteobacteria* and for three mobilizable transposons (*Bacteroidetes* and *Firmicutes*) as indicated in Table S7. Transposon CTnBST was also demonstrated to be conjugative (Gupta *et al.*, 2003). Plasmids pVEIS01, pA and pCC7120 α are assumed to be conjugative due to the presence of T4CPs and VirB4-like genes, although they are located far from the corresponding relaxase. Plasmid pCC7120 α is the first and only plasmid in the phylum *Cyanobacteria* for which there is experimental evidence of conjugative transfer (Muro-Pastor *et al.*, 1994).

The main well-resolved MOB_V clades are outlined in different colours in Fig. 16. Clade MOB_{V1} is the most populated. It comprises relaxases encoded by *Firmicutes* plasmids showing different degrees of sequence conservation. The prototype is the streptococcal plasmid pMV158. Clade MOB_{V2} contains plasmids from *Proteobacteria* and

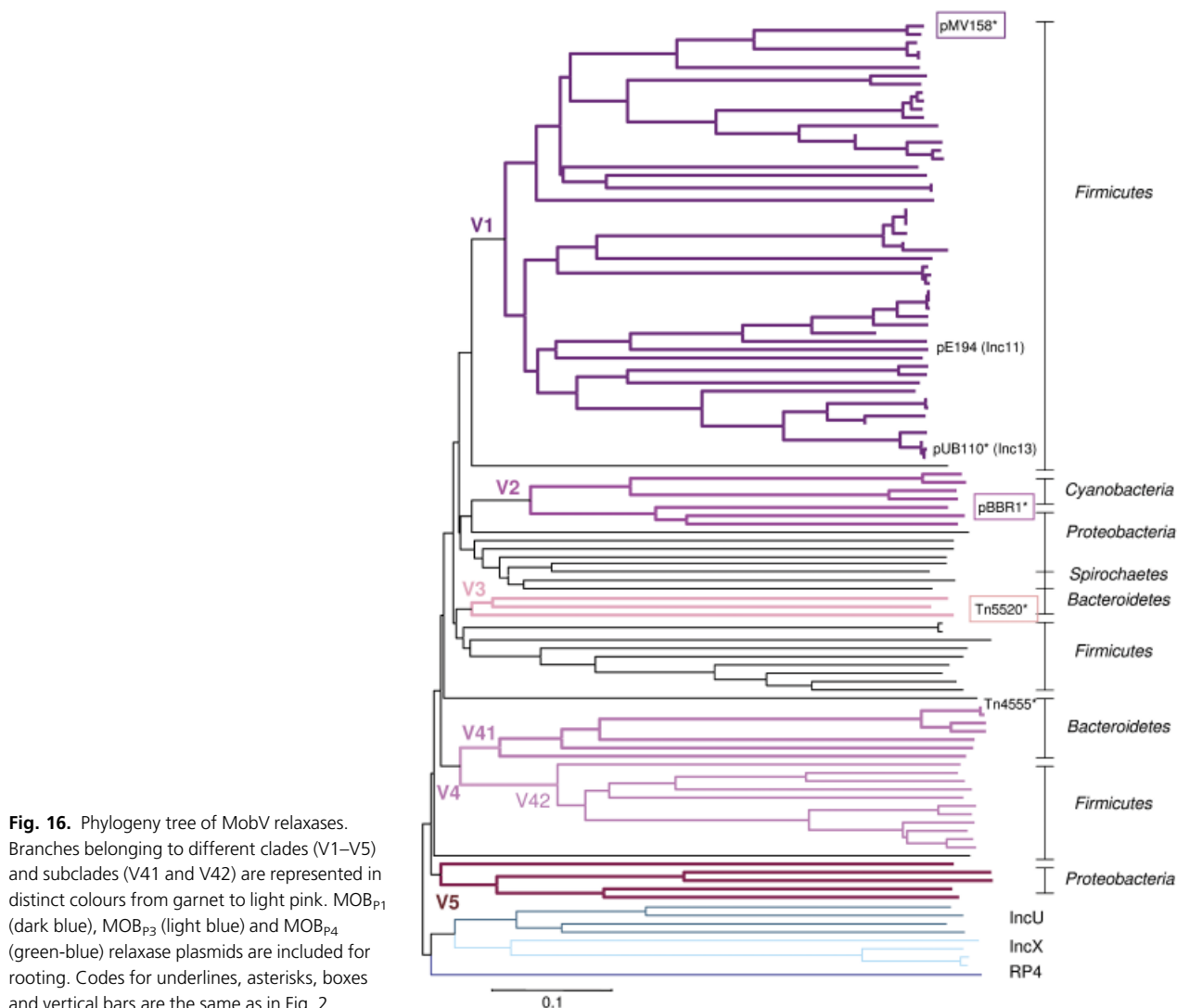
Cyanobacteria. The prototype is the θ -replicating plasmid pBBR1. Clade MOB_{V3} is a small clade including genetic elements from *Bacteroidetes*, of which relaxase in Tn5520 is the prototype. Clade MOB_{V4} contains plasmids and mobilizable transposons from *Bacteroidetes* and *Firmicutes*. Tn4555 is considered the prototype. Other clades indicated in the figure are small and composed of nonstudied plasmids.

Although MOB_V relaxases were first believed to be associated exclusively to RCR plasmids, θ -replicating plasmids as well as mobilizable and conjugative transposons do also contain MOB_V relaxases. Although the family has grown considerably in the last years, Motif I (H-x₂-R) and Motif III (H-x-DE-x₂-PH-x-H), as defined in Francia *et al.* (2004), are still generally well conserved (Fig. 17). Major exceptions belong to *Bacteroidetes* relaxases in clades MOB_{V3} and MOB_{V4}. Motif I in these relaxases shows the signature (HI)-x₂-R-x₂-E while *Bacteroidetes* relaxases from the genetic elements in Clade MOB_{V4} contain an extended Motif III (H-x-DE-x₈₋₂₄-P-x₂-H-x-H). Clade MOB_{V5} is the most deviant and Motif I is not found at all in its members. Nevertheless, PSI-BLAST searches started from pW240, pRm1132, pPMA4326D or pC converged with MOB_V relaxases in all cases. Thus, we believe they are bona fide members of the MOB_V family.

Unfortunately, we have very little data on the characterization of MOB_V relaxases. The MOB_{V1} relaxase MobM_pMV158 was purified by de Antonio *et al.* (2004) and shown to behave *grosso modo* as a classical relaxase. Specific recognition and nicking of its cognate *oriT* has also been demonstrated for the MOB_{V2} relaxase Mob_pBBR1 (Szpirer *et al.*, 2001). The interaction of the MOB_{V3} relaxase BmpH_Tn5520 with its *oriT* has also been analysed (Vedantam *et al.*, 2006). Nevertheless, when looking at the conserved motifs in MOB_V relaxases, it seems obvious that they are quite different from MOB_P or MOB_F relaxases, although it seems that the 3H motif III is conserved (Fig. 17). Besides, mutagenesis experiments on the MOB_{V2} relaxase Mob_pBBR1 showed that residues D120 and E121 (invariant residues in motif III) were essential for function (Szpirer *et al.*, 2001). Strikingly, the same authors mutated all seven Tyr residues in Mob_pBBR1 but none of them seemed required for function. This fact invokes a radically different mode of action for MOB_V relaxases. Thus, until more work is conducted, we are still doubtful about the DNA-processing mechanism of this family of relaxases.

Other relaxases: Tn916 relaxase

Orf20_Tn916, the relaxase encoded by the enterococcal 18-kb conjugative transposon Tn916, is a 329-amino acid protein, which holds similarity to neither of the MOB families described in this work. Nevertheless, it cleaves Tn916 *oriT* DNA and remains covalently associated to it (Rocco & Churchward, 2006). Tn916 integrase acts as an accessory



protein in the nicking reaction. It acts as a specificity determinant by binding to *oriT* just upstream of the cleavage site and giving cleavage specificity to Orf20 nuclease activity. A PSI-BLAST search using Orf20_Tn916 as a query retrieved proteins that belong to the Rep_trans superfamily of replication initiation factors, which includes RCR initiators from plasmids and phages (pfam02486, COG2946 and PHA00202 in NCBI database) but resulted in no hits against known relaxases. Orf20 homologues are present in a number of well-known ICEs from Gram-positive bacteria, such as Tn5397 (*Clostridium difficile*) (Roberts *et al.*, 2001), CW459tet(M) (*Clostridium perfringens*) (Roberts *et al.*, 2001), Tn5251 (*Streptococcus pneumoniae*) (Provvedi *et al.*, 1996), Tn5386 (*E. faecium*) (Rice *et al.*, 2007), Tn5801 (*S. aureus*) (Kuroda *et al.*, 2001), ICEst1 (*Streptococcus thermophilus*) (Burrus *et al.*, 2002), ICEBs1 (*Bacillus subtilis*) (Burrus *et al.*, 2002) and ICELm1 (*Listeria monocytogenes*) (Burrus *et al.*, 2002).

The high prevalence of these Tn916-like elements among many Gram-positive species and their association to a number of antibiotic resistance genes suggests that they play an essential role in the spread of antibiotic resistance. Analogues of RCR motifs II and III (that contain the histidine residues responsible of Mg²⁺ coordination and the tyrosine catalytic residue, respectively) were identified in Orf20_Tn916 (Rocco & Churchward, 2006). However, the authors propose a different origin for Orf20 and its closely related proteins from that of the RCR and relaxase proteins due to the observed motif differences. Thus, the status of Orf20_Tn916 remains unsettled. This could be an example of new relaxase families that remain to be characterized.

Related to Tn916 is conjugation system of *C. perfringens* plasmid pCW3 (Bannam *et al.*, 2006). Although pCW3 contains an 11-gene conjugation locus, which includes a T4CP (Parsons *et al.*, 2007), no relaxase-like gene was

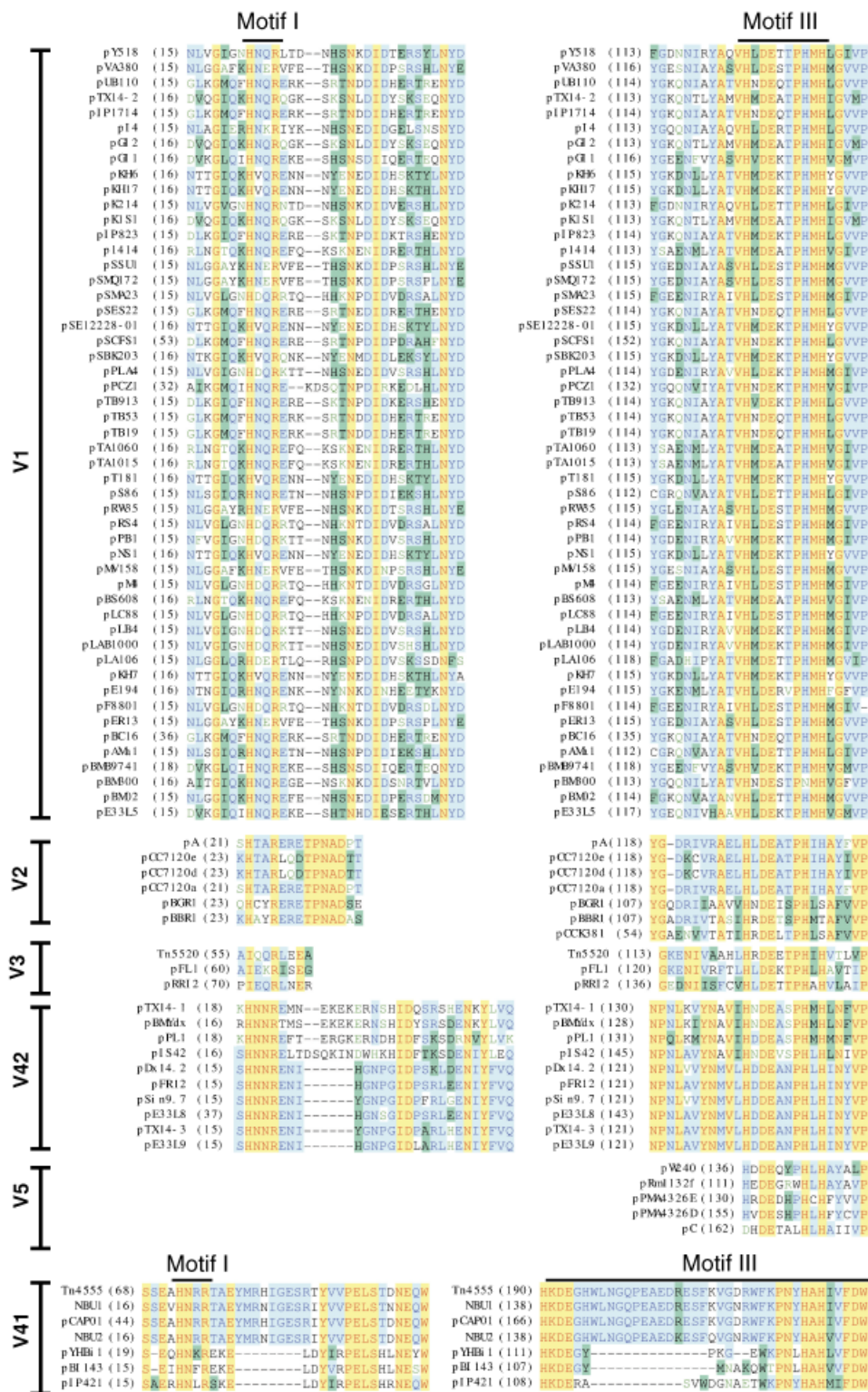


Fig. 17. Conserved sequence motifs in the MOB_V family. Codes are the same as in Fig. 3.

identified. Besides the *Clostridium* plasmids, some *Streptomyces* plasmids (Grohmann *et al.*, 2003) and archaeal plasmids (Greve *et al.*, 2004) have been reported in which no relaxase was identified. Because these potentially conjugative systems, as well as their bacterial hosts, are still in the early stages of analysis, we should not draw conclusions from them until some molecular details of the biochemical mechanisms involved become available.

Relaxase sequences as tools for the classification of conjugative systems

Complete plasmid sequencing and subsequent whole-genome comparison is the best criterion for plasmid classification. But this is obviously not attainable or even desirable when many samples have to be processed in the short term, when samples may include many repetitions of the same plasmid, etc. It is for this reason that specific sequences are usually selected as representatives of complete genomes and used as express classification tools. The great success in the use of 16S rRNA gene sequences for the classification of living organisms (Woese *et al.*, 1990) attests on the validity of this point of view. In this vein, plasmids are now routinely classified by the molecular characteristics of their replicons. This technique, called replicon typing (Couturier *et al.*, 1988; Carattoli *et al.*, 2005), has also been quite successful although it has shown several drawbacks. The first and most important caveat was that the set of sequences used for replicon typing was limited essentially to the classical enterobacterial incompatibility groups. A second crippling problem was that many plasmids, especially large plasmids, contain several replicons, making classification difficult. For example, IncFI plasmids harbour up to three replicons, some of them exhibiting recombination (Bergquist *et al.*, 1986; Couturier *et al.*, 1988); IncX2 plasmids contain a single replicase protein, but three different origins of replication (Crosa *et al.*, 1976; Shon *et al.*, 1982; Miron *et al.*, 1992). A third problematic issue was that several different replication systems exist, some requiring protein initiators of either RC-replication or θ -replication, others using not protein but RNA primers, etc. Thus, on many occasions, replication regions are difficult to identify even in completely sequenced plasmids. Plasmid classification using relaxases overcomes these three problems.

First, and most importantly, MOB phylogeny trees cover the whole bacterial diversity, at least for those bacteria in which plasmids have been described and sequenced. The classification of relaxases is most complete for the phylum *Proteobacteria*, which contains roughly 60% of the sequenced plasmids. Second in the rank come the *Firmicutes*, which contain an additional 25% of the sequenced plasmids. For the remaining phyla, the number of sequences analysed is still clearly insufficient. However, with the newly available

high-throughput sequencing techniques, the number of plasmid sequences is bound to increase exponentially in the next five years. We hope our classification scheme serves to put some order in the avalanche of this new information and as a tool for the researchers concerned in the classification of mobile elements and their constituent gene products [i.e. ACLAME database (<http://aclame.ulb.ac.be/>)] (Leplae *et al.*, 2004).

Second, plasmids rarely code for more than one relaxase. The reasons for this fact are not completely evident, but may have to do with the mechanism of conjugative DNA-processing itself. The survival of a plasmid DNA molecule may be compromised if more than one conjugative DNA processing event starts at the same time. The fact is that, in the list of over 600 plasmids we used, if we exclude the *Agrobacterium* plasmids (that contain a conjugative MOB_Q system together with a MOB_{P2} system for T-DNA transport to plant cells) only 15 plasmids contain two relaxases (conjugation was tested in none of them). In several of these plasmids, the situation might be similar to that of *Agrobacterium*. For instance, plasmid p42a from *Rhizobium etli* contains a MOB_Q-type TraA relaxase as part of a complete Tra genetic system, together with a MOB_{P2}-like VirD2 relaxase as part of a complete Vir genetic system although there is no obvious T-DNA associated to it (Gonzalez *et al.*, 2006). Although *Rhizobium* is not known to transport DNA to plant cells, one wonders what the function of the Vir system in these bacteria is. As a second example of the origin of bi-relaxase plasmids, we have the formation of cointegrates, a phenomenon that sometimes occurs under selective pressure. For instance, the enterococcal plasmid pAM α 1 contains genes almost identical to those in the tetracycline-resistance *Bacillus* RCR plasmid pBC16 plus those of pS86, a cryptic plasmid from *E. faecalis*. In the absence of tetracycline, the cointegrate resolves in *E. faecalis* cells and plasmid pS86 is mostly maintained. However, in the presence of the antibiotic, pAM α 1 generates amplification of the tetracycline resistance determinant. These processes occur via site-specific recombination involving pAM α 1 relaxases, MobB and MobE, and their *oriT* sites (Francia & Clewell, 2002a). Both relaxases belong to the MOB_V superfamily. We suspect other bi-relaxase plasmids may have analogous causal explanations.

Third, relaxase proteins are now easily identified in all plasmids capable of conjugation. We do not know of any conjugative plasmid that is not contained in one or another MOB family described in this review (but see Other relaxases: Tn916 relaxase). Although mechanisms of conjugation may differ in substantial ways among the six relaxase families, as discussed in this review, the six families together embrace all the diversity available in the databases. We hesitated and discarded to give easy PROSITE signatures to identify relaxases, because BLAST alignments are easy

enough to run and will univocally assign any relaxase to one of the MOB families.

An additional problematic issue in relation to plasmid classification is a consequence of the presumed modular evolution of plasmids (Novick, 1989; Berg *et al.*, 1998; Osborn *et al.*, 2000). We showed in this review (Figs 4 and S1–S4) the congruence between the relaxase and the T4CP phylogeny trees for the MOB_F, MOB_H, MOB_C, MOB_Q and MOB_{P1} families. It is clear from these phylogenies that relaxases and T4CPs evolve congruently for long periods of time. In other words, MOB regions are stable evolutionary units. The coevolution of MOB regions and T4SSs was already observed and analysed for the MOB_F family of plasmids (Fernandez-Lopez *et al.*, 2006). Although MOB and T4SS modules recombine more often, congruence between MOB and T4SS trees is still high unless we are dealing with very deep branching clades. Occasionally, however, MOB and T4SS regions swap between plasmids. This suggests that different MOB and T4SS can rearrange and adapt to each other so that they can reformulate a new transfer system. In fact, experimentally, it is easier to swap complete T4SSs than their individual components (Bolland *et al.*, 1990 and our unpublished data). In general, we infer from the data analysed here and elsewhere that relaxases rightly represent the whole transfer machinery for considerably broad clades.

Phylogenetic trees of a slowly changing protein as a relaxase, that can be considered a molecular clock, are a better representation of diversity than the Inc groups, which depend on the interaction of replication and stability proteins and can change with a single mutation. The trees give approximate evolutionary distances between branches. Thus, we see that the IncF, IncI or IncP complexes group relaxases with phylogenetic distances up to 0.4, while other groups, such as IncW or IncN, are probably more compact phylogenetically, with members presently not separated for more than 0.05 distance units.

Although the Inc notation is archaic, it has not yet been substituted. We do not propose to do it here, because many of the clades and subclades we proposed are still clearly underdetermined because there are few sequences available. But with the new generation sequencing strategies and the expected sequencing of many plasmid-rich metagenomic samples, the amount of relaxase sequences is expected to grow exponentially. The MOB notation that we introduce in the phylogenetic trees shown in this review can be expanded when new information becomes available without invalidating the previous assignments.

The main relaxase families, clades and subclades, and their prototype plasmids are shown in Table 1. As can be inferred from the table, most classical Inc groups of plasmids (specifically those of *Enterobacteriaceae*) are now located in the relaxase phylogenies. They could be used as evolutionary markers to define the place of a plasmid in the

universe of plasmid diversity. When two relaxases are > 90% identical, there is an almost certainty that the corresponding plasmids will share the complete backbone, as discussed before for IncW plasmids (Revilla *et al.*, 2008). Identification of the plasmid backbone should be the main use of classification by relaxases. A finer classification, sometimes necessary for epidemiological analysis, might need additional detail, such as multilocus typing (García-Fernández *et al.*, 2008) or even whole plasmid sequencing. Finally, it should be stressed that classification by relaxases is not meant to substitute replicon typing, but to complement it and provide additional information. As shown in Table 1, the classical Inc groups can also be named by one of the branches of the relaxase classification. This allows researchers to properly locate their test plasmid within the overall spread of plasmid diversity.

Concluding remarks

The subject of this review is the classification of relaxase proteins. Relaxases are essential proteins in plasmid conjugation. Because they start and end the DNA-processing reactions, they are decisive in the conjugation pathway. High relaxase sequence similarity is then synonymous to identical conjugative DNA-processing mechanism. Because evolution of relaxases and T4CPs is highly congruent, this review is also about classification of MOB regions, that is, DNA processing mechanisms during conjugation. Thus, MOB genetic systems can be classified in six families, as shown schematically in Fig. 1. Each family is somehow specific in the details of its DNA-processing mechanism. The MOB_F family forms a coherent, well-resolved group of plasmids. MOB_F relaxases contain two Tyr in their catalytic centre and mostly occur in conjugative plasmids. The MOB_P/MOB_Q/MOB_{HEN}/MOB_V cluster unites a large number of MOB regions whose relaxases contain just one tyrosine in the active centre (with a caveat in the case of MOB_V, according to the mutagenesis experiments on pBBR1). In spite of these differences, the mechanism of DNA processing in the MOB_P cluster seems to be similar to MOB_F, because the 3D structures of MOB_F relaxases (TrwC_R388 and TraI_F) and that of the MOB_{Q1} member MobA_RSF1010 are almost identical (see The MOB_Q family). In fact, MOB_F, MOB_Q and MOB_P show conservation of three motifs that configure the protein catalytic centre and are related to the relaxase mechanism of action, as shown in Figs 3, 10 and 13, the most conspicuous being the 3H motif. We propose to call this major group of relaxases the 3H class (defined by the existence of the Histidine triad in the catalytic centre). Judging from the point of view of relaxase classification, the present knowledge examined in this review indicates that 3H-relaxases containing MOB systems constitute the

Table 1. Representative relaxase protein clades

Plasmid family	Clade	Prototype relaxase_plasmid (incompatibility group)*	Representative references
MOB _F	F11	TrwC_R388 (IncW)	Gonzalez-Perez <i>et al.</i> (2007)
		Tral_R46 (IncN)	Paterson <i>et al.</i> (1999)
	F12	pWWO (IncP9)	Greated <i>et al.</i> (2002)
		Tral_F (IncFI)	Hekman <i>et al.</i> (2008)
		Tral_R1 (IncFII)	Csitkovits <i>et al.</i> (2004)
MOB _H	H11	Tral_pED208 (IncFV)	Lu <i>et al.</i> (2002)
		Tral_R27 (IncHI1)	Lawley <i>et al.</i> (2002)
	H12	Tral_R478 (IncHI2)	Gilmour <i>et al.</i> (2004)
		Tral_pIP1202 (IncA/C)	Welch <i>et al.</i> (2007)
		Tral_R391 (IncJ)	Boltner & Osborn (2004)
	H2	Tral_Rts1 (IncT)	Murata <i>et al.</i> (2002)
		Tra_L_pCAR1 (IncP7)	Maeda <i>et al.</i> (2003)
		Tral_Neigo (chromosomal GI)	Salgado-Pabon <i>et al.</i> (2007)
MOB _C	C1	MobC_CloDF13 (mobilizable)	Nunez & de la Cruz (2001)
	C2	TraX_pAD1	Francia & Clewell (2002b)
MOB _Q	Q1	MobA_RSF1010 (IncQ1/IncP4)	Parker & Meyer (2007)
	Q2	TrA_pTiC58	Cho & Winans (2007)
MOB _P		TrA_p42d	Perez-Mendoza <i>et al.</i> (2006)
	Q3	TrA_pIP501 (Inc18)	Abajy <i>et al.</i> (2007)
	P11	Tral_RP4 (IncP1α)	Pansegrau & Lanka (1996)
		Tral_R751 (IncP1β)	Thorsted <i>et al.</i> (1998)
	P12	NikB_R64 (IncI1)	Furuya <i>et al.</i> (1991)
		NikB_R387 (IncK)	Tschape & Tietze (1980)
	P13	pCTX-M3 (IncL/M)	Golebiewski <i>et al.</i> (2007)
	P14	MobA_pTF-FC2 (IncQ2α) (mobilizable)	van Zyl <i>et al.</i> (2003)
		MobA_pTC-F14 (IncQ2β) (mobilizable)	van Zyl <i>et al.</i> (2003)
		MobA_Rms149 (IncG/IncP6)	Haines <i>et al.</i> (2005)
	P2	VirD2_pTiC58 (T-DNA transport)	Scheiffele <i>et al.</i> (1995)
	P3	TaxC_pOLA52 (IncX1)	Norman <i>et al.</i> (2008)
		TaxC_R6K (IncX2)	Nunez <i>et al.</i> (1997)
	P4	VirD2_pFBAOT6 (IncU)	Rhodes <i>et al.</i> (2004)
		TrAS_pSB102	Schneiker <i>et al.</i> (2001)
	P5 (MOB _{HEN})	MbeA_ColE1 (mobilizable)	Varsaki <i>et al.</i> (2003)
P6	NikB_R721 (IncI2)	Komano <i>et al.</i> (1990)	
MOB _V	P7	PcfG_pCF10	Chen <i>et al.</i> (2007)
		MobA_pC221 (Inc4)	Caryl & Thomas (2006)
	V1	MobA_pC223 (Inc10)	Caryl <i>et al.</i> (2004)
		MobM_pMV158	de Antonio <i>et al.</i> (2004)
		pE194 (Inc11)	Horinouchi & Weisblum (1982)
	V2	pUB110 (Inc13)	Lotareva <i>et al.</i> (2001)
		Mob_pBBR1	Szipirer <i>et al.</i> (2001)
	V3	BmpH_Tn5520 (chromosomal Tn)	Vedantam <i>et al.</i> (2006)
	V4	MobA_Tn4555 (chromosomal Tn)	Smith & Parker (1998)
Unclassified		Orf20_Tn916 (chromosomal Tn)	Rocco & Churchward (2006)

*Incompatibility groups were described for plasmids in *Enterobacteria*, *Pseudomonas* and *S. aureus*. They were abandoned as a classification method as explained in the introduction. IncD exists (Coetzee *et al.*, 1985), but no member has been sequenced. IncY is composed of plasmids related to phage P1 (Capage *et al.*, 1982). There are no references for IncE, IncR, IncS or IncV.

predominant mechanism of plasmid conjugation. The remaining MOB groups are clearly deviant from this mechanism. The MOB_H family is composed of another coherent and well-resolved group of proteins. Although two tyrosines have been invoked as participating in DNA processing, these relaxases are clearly different from the previous groups in

significant aspects (see The MOB_H family). It might be that the mechanism of DNA processing of MOB_H systems is different from that of 3H relaxases, although it is not yet certain. Finally, the MOB_C family is the most remotely related of all. We are not sure whether there is a Tyr in their catalytic centres or, even, how is the mechanism of DNA

processing as the relaxase-cleaved DNA does not remain covalently bound to the protein, but shows free 3'-ends (see The MOB_C family). Further biochemical analysis of MOB_C systems is required to solve these important questions.

Conjugative relaxases have remote homologues in other phosphodiesterase protein families with different functions. For instance, MOB_C relaxases are related to RC-replication (Rep) proteins (see The MOB_C family), MOB_F relaxases to RC-transposition proteins (see The MOB_F family) and MOB_H relaxases to HD hydrolases (see The MOB_H family). This implies that the phenomena of conjugation, replication and transposition are evolutionarily connected, at least in some of their variants. This represents to us striking examples of the plasticity of the various relaxase protein folds, which can be adapted to widely different functions. Alternatively, conjugation systems evolved by recruitment of phosphodiesterases that were performing other unrelated functions.

In conclusion, relaxases might be useful to classify plasmids (besides conjugative systems themselves). We can learn much about the properties of plasmids just by sequencing a 1-kb sequence containing the relaxase. The prevalence of a given MOB system provides information about the ecological preferences of some plasmid types. We do not know enough about mechanisms to explain these differences. Each MOB type contains details of plasmid idiosyncrasy that need to be explored further. We hope our work can be useful in this context, because it provides phylogenetic as well as biochemical data that group plasmids according to a functionally important property, and thus probably reflect outstanding functional decisions in the evolution of plasmid sequences. It is also expected that our effort in classifying the actual diversity of relaxases (and hence of plasmid diversity) will help us to reach a better understanding of the physiology of plasmids and of their conjugation mechanisms.

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References

- Abajy MY, Kopec J, Schiwon K, Burzynski M, Doring M, Bohn C & Grohmann E (2007) A type IV-secretion-like system is required for conjugative DNA transport of broad-host-range plasmid pIP501 in gram-positive bacteria. *J Bacteriol* **189**: 2487–2496.
- Alonso G, Baptista K, Ngo T & Taylor DE (2005) Transcriptional organization of the temperature-sensitive transfer system from the IncHI1 plasmid R27. *Microbiology* **151**: 3563–3573.
- Altschul SF, Madden TL, Schaffer AA, Zhang J, Zhang Z, Miller W & Lipman DJ (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res* **25**: 3389–3402.
- Aravind L & Koonin EV (1998) The HD domain defines a new superfamily of metal-dependent phosphohydrolases. *Trends Biochem Sci* **23**: 469–472.
- Avila P, Nunez B & de la Cruz F (1996) Plasmid R6K contains two functional oriTs which can assemble simultaneously in relaxosomes *in vivo*. *J Mol Biol* **261**: 135–143.
- Bahl MI, Hansen LH, Goesmann A & Sorensen SJ (2007) The multiple antibiotic resistance IncP-1 plasmid pKJK5 isolated from a soil environment is phylogenetically divergent from members of the previously established alpha, beta and delta sub-groups. *Plasmid* **58**: 31–43.
- Bannam TL, Teng WL, Bulach D, Lyras D & Rood JI (2006) Functional identification of conjugation and replication regions of the tetracycline resistance plasmid pCW3 from *Clostridium perfringens*. *J Bacteriol* **188**: 4942–4951.
- Beaber JW, Burrus V, Hochhut B & Waldor MK (2002) Comparison of SXT and R391, two conjugative integrating elements: definition of a genetic backbone for the mobilization of resistance determinants. *Cell Mol Life Sci* **59**: 2065–2070.
- Berg T, Firth N, Apisiridej S, Hettiaratchi A, Leelaporn A & Skurray RA (1998) Complete nucleotide sequence of pSK41: evolution of staphylococcal conjugative multiresistance plasmids. *J Bacteriol* **180**: 4350–4359.
- Bergquist PL, Saadi S & Maas WK (1986) Distribution of basic replicons having homology with RepFIA, RepFIB, and RepFIC among IncF group plasmids. *Plasmid* **15**: 19–34.
- Bolland S, Llosa M, Avila P & de la Cruz F (1990) General organization of the conjugal transfer genes of the IncW plasmid R388 and interactions between R388 and IncN and IncP plasmids. *J Bacteriol* **172**: 5795–5802.
- Boltner D & Osborn AM (2004) Structural comparison of the integrative and conjugative elements R391, pMERPH, R997, and SXT. *Plasmid* **51**: 12–23.
- Boltner D, MacMahon C, Pembroke JT, Strike P & Osborn AM (2002) R391: a conjugative integrating mosaic comprised of phage, plasmid, and transposon elements. *J Bacteriol* **184**: 5158–5169.
- Boyd EF, Hill CW, Rich SM & Hartl DL (1996) Mosaic structure of plasmids from natural populations of *Escherichia coli*. *Genetics* **143**: 1091–1100.
- Bradley DE (1984) Characteristics and function of thick and thin conjugative pili determined by transfer-derepressed plasmids

- of incompatibility groups I1, I2, I5, B, K and Z. *J Gen Microbiol* **130**: 1489–1502.
- Bradley DE, Taylor DE & Cohen DR (1980) Specification of surface mating systems among conjugative drug resistance plasmids in *Escherichia coli* K-12. *J Bacteriol* **143**: 1466–1470.
- Burrus V, Pavlovic G, Decaris B & Guedon G (2002) The ICEst1 element of *Streptococcus thermophilus* belongs to a large family of integrative and conjugative elements that exchange modules and change their specificity of integration. *Plasmid* **48**: 77–97.
- Capage MA, Goodspeed JK & Scott JR (1982) Incompatibility group Y member relationships: pIP231 and plasmid prophages P1 and P7. *Plasmid* **8**: 307–311.
- Carattoli A, Bertini A, Villa L, Falbo V, Hopkins KL & Threlfall EJ (2005) Identification of plasmids by PCR-based replicon typing. *J Microbiol Meth* **63**: 219–228.
- Caryl JA & Thomas CD (2006) Investigating the basis of substrate recognition in the pC221 relaxosome. *Mol Microbiol* **60**: 1302–1318.
- Caryl JA, Smith MC & Thomas CD (2004) Reconstitution of a staphylococcal plasmid-protein relaxation complex *in vitro*. *J Bacteriol* **186**: 3374–3383.
- Cattoir V, Poirel L, Aubert C, Soussy CJ & Nordmann P (2008) Unexpected occurrence of plasmid-mediated quinolone resistance determinants in environmental *Aeromonas* spp. *Emerg Infect Dis* **14**: 231–237.
- Cesar CE, Machon C, de la Cruz F & Llosa M (2006) A new domain of conjugative relaxase TrwC responsible for efficient oriT-specific recombination on minimal target sequences. *Mol Microbiol* **62**: 984–996.
- Chen Y, Staddon JH & Dunne GM (2007) Specificity determinants of conjugative DNA processing in the *Enterococcus faecalis* plasmid pCF10 and the *Lactococcus lactis* plasmid pRS01. *Mol Microbiol* **63**: 1549–1564.
- Chilley PM & Wilkins BM (1995) Distribution of the *ardA* family of antirestriction genes on conjugative plasmids. *Microbiology* **141**: 2157–2164.
- Cho H & Winans SC (2007) TraA, TraC and TraD autorepress two divergent quorum-regulated promoters near the transfer origin of the Ti plasmid of *Agrobacterium tumefaciens*. *Mol Microbiol* **63**: 1769–1782.
- Christie PJ (2004) Type IV secretion: the *Agrobacterium* VirB/D4 and related conjugation systems. *Biochim Biophys Acta* **1694**: 219–234.
- Clewell DB (2007) Properties of *Enterococcus faecalis* plasmid pAD1, a member of a widely disseminated family of pheromone-responding, conjugative, virulence elements encoding cytolysin. *Plasmid* **58**: 205–227.
- Cloekaert A, Praud K, Doublet B *et al.* (2007) Dissemination of an extended-spectrum-beta-lactamase blaTEM-52 gene-carrying IncI1 plasmid in various *Salmonella enterica* serovars isolated from poultry and humans in Belgium and France between 2001 and 2005. *Antimicrob Agents Ch* **51**: 1872–1875.
- Coburn PS, Baghdadyan AS, Dolan GT & Shankar N (2007) Horizontal transfer of virulence genes encoded on the *Enterococcus faecalis* pathogenicity island. *Mol Microbiol* **63**: 530–544.
- Coetsee JN, Datta N & Hedges RW (1972) R factors from *Proteus rettgeri*. *J Gen Microbiol* **72**: 543–552.
- Coetsee JN, Bradley DE, Lecatsas G, du Toit L & Hedges RW (1985) Bacteriophage D: an IncD group plasmid-specific phage. *J Gen Microbiol* **131**: 3375–3383.
- Coque TM, Novais A, Carattoli A *et al.* (2008) Dissemination of clonally related *Escherichia coli* strains expressing extended-spectrum beta-lactamase CTX-M-15. *Emerg Infect Dis* **14**: 195–200.
- Couturier M, Bex F, Bergquist PL & Maas WK (1988) Identification and classification of bacterial plasmids. *Microbiol Rev* **52**: 375–395.
- Crosa JH, Luttrupp LK & Falkow S (1976) Mode of replication of the conjugative R-plasmid RSF1040 in *Escherichia coli*. *J Bacteriol* **126**: 454–466.
- Csirikovits VC, Ethermic D & Zechner EL (2004) Concomitant reconstitution of TraI-catalyzed DNA transesterase and DNA helicase activity *in vitro*. *J Biol Chem* **279**: 45477–45484.
- de Antonio C, Farias ME, de Lacoba MG & Espinosa M (2004) Features of the plasmid pMV158-encoded MobM, a protein involved in its mobilization. *J Mol Biol* **335**: 733–743.
- De Boever EH, Clewell DB & Fraser CM (2000) *Enterococcus faecalis* conjugative plasmid pAM373: complete nucleotide sequence and genetic analyses of sex pheromone response. *Mol Microbiol* **37**: 1327–1341.
- de la Cruz F, Zabala JC & Ortiz JM (1979) Incompatibility among alpha-hemolytic plasmids studied after inactivation of the alpha-hemolysin gene by transposition of Tn802. *Plasmid* **2**: 507–519.
- Dudley EG, Abe C, Ghigo JM, Latour-Lambert P, Hormazabal JC & Nataro JP (2006) An IncI1 plasmid contributes to the adherence of the atypical enteroaggregative *Escherichia coli* strain C1096 to cultured cells and abiotic surfaces. *Infect Immun* **74**: 2102–2114.
- Evans TJ, Pérez-Mendoza D, Monson RE, Stickland HG & Salmond GPC (2009). Secretion systems of the enterobacterial phytopathogen, *Erwinia* (K. Wooldridge, ed.) Caister Academic Press, Nottingham, UK.
- Fernandez-Lopez R, Garcillan-Barcia MP, Revilla C, Lazaro M, Vielva L & de la Cruz F (2006) Dynamics of the IncW genetic backbone imply general trends in conjugative plasmid evolution. *FEMS Microbiol Rev* **30**: 942–966.
- Francia MV & Clewell DB (2002a) Amplification of the tetracycline resistance determinant of pAMalpha1 in *Enterococcus faecalis* requires a site-specific recombination event involving relaxase. *J Bacteriol* **184**: 5187–5193.
- Francia MV & Clewell DB (2002b) Transfer origins in the conjugative *Enterococcus faecalis* plasmids pAD1 and pAM373: identification of the pAD1 *nic* site, a specific relaxase and a possible TraG-like protein. *Mol Microbiol* **45**: 375–395.
- Francia MV, Varsaki A, Garcillan-Barcia MP, Latorre A, Drains C & de la Cruz F (2004) A classification scheme for mobilization regions of bacterial plasmids. *FEMS Microbiol Rev* **28**: 79–100.

- Freitas AR, Francia MV, Peixe L, Novais C, Jensen L, Willems R, Baquero F & Coque TM (2008) *Plasmid characterization of vancomycin resistant Enterococcus faecium strains from different continents (1986–2007)*. 18th European Congress of Clinical Microbiology and Infectious Diseases. April 2008. Barcelona, Spain.
- Furuya N & Komano T (2003) NikAB- or NikB-dependent intracellular recombination between tandemly repeated oriT sequences of plasmid R64 in plasmid or single-stranded phage vectors. *J Bacteriol* **185**: 3871–3877.
- Furuya N, Nisioka T & Komano T (1991) Nucleotide sequence and functions of the oriT operon in IncI1 plasmid R64. *J Bacteriol* **173**: 2231–2237.
- García-Fernández A, Chiaretto G, Bertini A, Villa L, Fortini D, Ricci A & Carattoli A (2008) Multilocus sequence typing of IncI1 plasmids carrying extended-spectrum beta-lactamases in *Escherichia coli* and *Salmonella* of human and animal origin. *J Antimicrob Chemother* **61**: 1229–1233.
- Garcillán-Barcia MP & de la Cruz F (2002) Distribution of IS91 family insertion sequences in bacterial genomes: evolutionary implications. *FEMS Microbiol Ecol* **42**: 303–313.
- Garcillán-Barcia MP, Bernales I, Mendiola MV & de la Cruz F (2002) IS91 rolling-circle transposition. *Mobile DNA II* (Craig NL, ed.), pp. 891–904. ASM Press, Washington, DC.
- Gilmour MW, Thomson NR, Sanders M, Parkhill J & Taylor DE (2004) The complete nucleotide sequence of the resistance plasmid R478: defining the backbone components of incompatibility group H conjugative plasmids through comparative genomics. *Plasmid* **52**: 182–202.
- Golebiewski M, Kern-Zdanowicz I, Zienkiewicz M *et al.* (2007) Complete nucleotide sequence of the pCTX-M3 plasmid and its involvement in spread of the extended-spectrum beta-lactamase gene blaCTX-M-3. *Antimicrob Agents Ch* **51**: 3789–3795.
- Gonzalez V, Santamaria RI, Bustos P *et al.* (2006) The partitioned *Rhizobium etli* genome: genetic and metabolic redundancy in seven interacting replicons. *P Natl Acad Sci USA* **103**: 3834–3839.
- Gonzalez-Perez B, Lucas M, Cooke LA, Vyle JS, de la Cruz F & Moncalian G (2007) Analysis of DNA processing reactions in bacterial conjugation by using suicide oligonucleotides. *EMBO J* **26**: 3847–3857.
- Grandoso G, Avila P, Cayon A, Hernando MA, Llosa M & de la Cruz F (2000) Two active-site tyrosyl residues of protein TrwC act sequentially at the origin of transfer during plasmid R388 conjugation. *J Mol Biol* **295**: 1163–1172.
- Greated A, Lambertsen L, Williams PA & Thomas CM (2002) Complete sequence of the IncP-9 TOL plasmid pWW0 from *Pseudomonas putida*. *Environ Microbiol* **4**: 856–871.
- Greve B, Jensen S, Brugger K, Zillig W & Garrett RA (2004) Genomic comparison of archaeal conjugative plasmids from *Sulfolobus*. *Archaea* **1**: 231–239.
- Grohmann E, Muth G & Espinosa M (2003) Conjugative plasmid transfer in gram-positive bacteria. *Microbiol Mol Biol R* **67**: 277–301.
- Guasch A, Lucas M, Moncalian G *et al.* (2003) Recognition and processing of the origin of transfer DNA by conjugative relaxase TrwC. *Nat Struct Biol* **10**: 1002–1010.
- Guindon S & Gascuel O (2003) A simple, fast, and accurate algorithm to estimate large phylogenies by maximum likelihood. *Syst Biol* **52**: 696–704.
- Guindon S, Lethiec F, Duroux P & Gascuel O (2005) PHYML Online – a web server for fast maximum likelihood-based phylogenetic inference. *Nucleic Acids Res* **33**: W557–W559.
- Gupta A, Vlamakis H, Shoemaker N & Salyers AA (2003) A new *Bacteroides* conjugative transposon that carries an ermB gene. *Appl Environ Microbiol* **69**: 6455–6463.
- Guzman LM & Espinosa M (1997) The mobilization protein, MobM, of the streptococcal plasmid pMV158 specifically cleaves supercoiled DNA at the plasmid oriT. *J Mol Biol* **266**: 688–702.
- Haines AS, Jones K, Cheung M & Thomas CM (2005) The IncP-6 plasmid Rms149 consists of a small mobilizable backbone with multiple large insertions. *J Bacteriol* **187**: 4728–4738.
- Haines AS, Cheung M & Thomas CM (2006a) Evidence that IncG (IncP-6) and IncU plasmids form a single incompatibility group. *Plasmid* **55**: 210–215.
- Haines AS, Akhtar P, Stephens ER *et al.* (2006b) Plasmids from freshwater environments capable of IncQ retrotransfer are diverse and include pQKH54, a new IncP-1 subgroup archetype. *Microbiology* **152**: 2689–2701.
- Hamilton HL & Dillard JP (2006) Natural transformation of *Neisseria gonorrhoeae*: from DNA donation to homologous recombination. *Mol Microbiol* **59**: 376–385.
- Hamilton HL, Dominguez NM, Schwartz KJ, Hackett KT & Dillard JP (2005) *Neisseria gonorrhoeae* secretes chromosomal DNA via a novel type IV secretion system. *Mol Microbiol* **55**: 1704–1721.
- Hekman K, Guja K, Larkin C & Schildbach JF (2008) An intrastrand three-DNA-base interaction is a key specificity determinant of F transfer initiation and of F TraI relaxase DNA recognition and cleavage. *Nucleic Acids Res* **36**: 4565–4572.
- Herrero A, Mendoza MC, Rodicio R & Rodicio MR (2008) Characterization of pUO-StVR2, a virulence-resistance plasmid evolved from the pSLT virulence plasmid of *Salmonella enterica* serovar Typhimurium. *Antimicrob Agents Ch* **52**: 4514–4517.
- Heuer H, Szczepanowski R, Schneiker S, Puhler A, Top EM & Schluter A (2004) The complete sequences of plasmids pB2 and pB3 provide evidence for a recent ancestor of the IncP-1beta group without any accessory genes. *Microbiology* **150**: 3591–3599.
- Horinouchi S & Weisblum B (1982) Nucleotide sequence and functional map of pE194, a plasmid that specifies inducible resistance to macrolide, lincosamide, and streptogramin type B antibiotics. *J Bacteriol* **150**: 804–814.
- Jones CS, Osborne DJ & Stanley J (1993) Molecular comparison of the IncX plasmids allows division into IncX1 and IncX2 subgroups. *J Gen Microbiol* **139**: 735–741.

- Kim SR & Komano T (1992) Nucleotide sequence of the R721 shufflon. *J Bacteriol* **174**: 7053–7058.
- Klockgether J, Reva O, Larbig K & Tummler B (2004) Sequence analysis of the mobile genome island pKLC102 of *Pseudomonas aeruginosa* C. *J Bacteriol* **186**: 518–534.
- Komano T, Fujitani S, Funayama N, Kanno A & Sakuma K (1990) Physical and genetic analyses of IncI2 plasmid R721: evidence for the presence of shufflon. *Plasmid* **23**: 248–251.
- Kulinska A, Czeredys M, Hayes F & Jagura-Burdzy G (2008) Genomic and functional characterization of the modular broad-host-range RA3 plasmid, the archetype of the IncU group. *Appl Environ Microbiol* **74**: 4119–4132.
- Kumar S, Tamura K & Nei M (2004) MEGA3: integrated software for molecular evolutionary genetics analysis and sequence alignment. *Brief Bioinform* **5**: 150–163.
- Kurenbach B, Kopec J, Magdefrau M *et al.* (2006) The TraA relaxase autoregulates the putative type IV secretion-like system encoded by the broad-host-range *Streptococcus agalactiae* plasmid pIP501. *Microbiology* **152**: 637–645.
- Kuroda M, Ohta T, Uchiyama I *et al.* (2001) Whole genome sequencing of methicillin-resistant *Staphylococcus aureus*. *Lancet* **357**: 1225–1240.
- Larkin C, Datta S, Harley MJ, Anderson BJ, Ebie A, Hargreaves V & Schildbach JF (2005) Inter- and intramolecular determinants of the specificity of single-stranded DNA binding and cleavage by the F factor relaxase. *Structure* **13**: 1533–1544.
- Lawley TD, Gilmour MW, Gunton JE, Standeven LJ & Taylor DE (2002) Functional and mutational analysis of conjugative transfer region 1 (Tra1) from the IncHI1 plasmid R27. *J Bacteriol* **184**: 2173–2180.
- Lawley TD, Gilmour MW, Gunton JE, Tracz DM & Taylor DE (2003) Functional and mutational analysis of conjugative transfer region 2 (Tra2) from the IncHI1 plasmid R27. *J Bacteriol* **185**: 581–591.
- Leplae R, Hebrant A, Wodak SJ & Toussaint A (2004) ACLAME: a CLAssification of mobile genetic elements. *Nucleic Acids Res* **32**: D45–D49.
- Lin TL, Lee CZ, Hsieh PF, Tsai SF & Wang JT (2008) Characterization of integrative and conjugative element ICEKp1-associated genomic heterogeneity in a *Klebsiella pneumoniae* strain isolated from a primary liver abscess. *J Bacteriol* **190**: 515–526.
- Llosa M, Gomis-Ruth FX, Coll M & de la Cruz F (2002) Bacterial conjugation: a two-step mechanism for DNA transport. *Mol Microbiol* **45**: 1–8.
- Lotareva OV, Nezametdinova VZ, Fedorina EA, Poluektova EU, Titok MA & Prozorov AA (2001) High frequency conjugative mobilization in natural strains of *Bacillus subtilis*, bearing a large plasmid. *Genetika* **37**: 1598–1603.
- Lu J, Manchak J, Klimke W, Davidson C, Firth N, Skurray RA & Frost LS (2002) Analysis and characterization of the IncFV plasmid pED208 transfer region. *Plasmid* **48**: 24–37.
- Maeda K, Nojiri H, Shintani M, Yoshida T, Habe H & Omori T (2003) Complete nucleotide sequence of carbazole/dioxin-degrading plasmid pCAR1 in *Pseudomonas resinovorans* strain CA10 indicates its mosaicity and the presence of large catabolic transposon Tn4676. *J Mol Biol* **326**: 21–33.
- Matson SW & Ragonese H (2005) The F-plasmid TraI protein contains three functional domains required for conjugative DNA strand transfer. *J Bacteriol* **187**: 697–706.
- Miron A, Mukherjee S & Bastia D (1992) Activation of distant replication origins *in vivo* by DNA looping as revealed by a novel mutant form of an initiator protein defective in cooperativity at a distance. *EMBO J* **11**: 1205–1216.
- Monzingo AF, Ozburn A, Xia S, Meyer RJ & Robertus JD (2007) The structure of the minimal relaxase domain of MobA at 2.1 Å resolution. *J Mol Biol* **366**: 165–178.
- Murata T, Ohnishi M, Ara T *et al.* (2002) Complete nucleotide sequence of plasmid Rts1: implications for evolution of large plasmid genomes. *J Bacteriol* **184**: 3194–3202.
- Muro-Pastor AM, Kuritz T, Flores E, Herrero A & Wolk CP (1994) Transfer of a genetic marker from a megaplasmid of *Anabaena* sp. strain PCC 7120 to a megaplasmid of a different *Anabaena* strain. *J Bacteriol* **176**: 1093–1098.
- Navarro F, Mesa RJ, Miro E, Gomez L, Mirelis B & Coll P (2007) Evidence for convergent evolution of CTX-M-14 ESBL in *Escherichia coli* and its prevalence. *FEMS Microbiol Lett* **273**: 120–123.
- Norman A, Hansen LH, She Q & Sorensen SJ (2008) Nucleotide sequence of pOLA52: a conjugative IncX1 plasmid from *Escherichia coli* which enables biofilm formation and multidrug efflux. *Plasmid* **60**: 59–74.
- Novais A, Canton R, Moreira R, Peixe L, Baquero F & Coque TM (2007) Emergence and dissemination of *Enterobacteriaceae* isolates producing CTX-M-1-like enzymes in Spain are associated with IncFII (CTX-M-15) and broad-host-range (CTX-M-1, -3, and -32) plasmids. *Antimicrob Agents Ch* **51**: 796–799.
- Novick RP (1989) Staphylococcal plasmids and their replication [published erratum appears in *Annu Rev Microbiol* 1990;44:10]. *Annu Rev Microbiol* **43**: 537–565.
- Nunez B & de la Cruz F (2001) Two atypical mobilization proteins are involved in plasmid CloDF13 relaxation. *Mol Microbiol* **39**: 1088–1099.
- Nunez B, Avila P & de la Cruz F (1997) Genes involved in conjugative DNA processing of plasmid R6K. *Mol Microbiol* **24**: 1157–1168.
- Osborn AM, da Silva Tatley FM, Steyn LM, Pickup RW & Saunders JR (2000) Mosaic plasmids and mosaic replicons: evolutionary lessons from the analysis of genetic diversity in IncFII-related replicons. *Microbiology* **146**: 2267–2275.
- Pansegau W & Lanka E (1996) Mechanisms of initiation and termination reactions in conjugative DNA processing. Independence of tight substrate binding and catalytic activity of relaxase (TraI) of IncPalph plasmid RP4. *J Biol Chem* **271**: 13068–13076.
- Pansegau W, Schoumacher F, Hohn B & Lanka E (1993) Site-specific cleavage and joining of single-stranded DNA by VirD2

- protein of *Agrobacterium tumefaciens* Ti plasmids: analogy to bacterial conjugation. *P Natl Acad Sci USA* **90**: 11538–11542.
- Pansegrau W, Schroder W & Lanka E (1994) Concerted action of three distinct domains in the DNA cleaving-joining reaction catalyzed by relaxase (TraI) of conjugative plasmid RP4. *J Biol Chem* **269**: 2782–2789.
- Parker C & Meyer RJ (2007) The R1162 relaxase/primase contains two, type IV transport signals that require the small plasmid protein MobB. *Mol Microbiol* **66**: 252–261.
- Parsons JA, Bannam TL, Devenish RJ & Rood JI (2007) TcpA, an FtsK/SpoIIIE homolog, is essential for transfer of the conjugative plasmid pCW3 in *Clostridium perfringens*. *J Bacteriol* **189**: 7782–7790.
- Paterson ES, More MI, Pillay G et al. (1999) Genetic analysis of the mobilization and leading regions of the IncN plasmids pKM101 and pCU1. *J Bacteriol* **181**: 2572–2583.
- Perez-Mendoza D, Lucas M, Munoz S, Herrera-Cervera JA, Olivares J, de la Cruz F & Sanjuan J (2006) The relaxase of the *Rhizobium etli* symbiotic plasmid shows *nic* site cis-acting preference. *J Bacteriol* **188**: 7488–7499.
- Praszkier J, Wei T, Siemering K & Pittard J (1991) Comparative analysis of the replication regions of IncB, IncK, and IncZ plasmids. *J Bacteriol* **173**: 2393–2397.
- Priebe SD & Lacks SA (1989) Region of the streptococcal plasmid pMV158 required for conjugative mobilization. *J Bacteriol* **171**: 4778–4784.
- Provvedi R, Manganelli R & Pozzi G (1996) Characterization of conjugative transposon Tn5251 of *Streptococcus pneumoniae*. *FEMS Microbiol Lett* **135**: 231–236.
- Rawlings DE (2005) The evolution of pTF-FC2 and pTC-F14, two related plasmids of the IncQ-family. *Plasmid* **53**: 137–147.
- Revilla C, Garcillán-Barcia MP, Fernández-López R et al. (2008) Different pathways to acquiring resistance genes illustrated by the recent evolution of IncW plasmids. *Antimicrob Agents Ch* **52**: 1472–1480.
- Rhodes G, Parkhill J, Bird C et al. (2004) Complete nucleotide sequence of the conjugative tetracycline resistance plasmid pFBAOT6, a member of a group of IncU plasmids with global ubiquity. *Appl Environ Microbiol* **70**: 7497–7510.
- Rice LB, Carias LL, Marshall SH, Hutton-Thomas R & Rudin S (2007) Characterization of Tn5386, a Tn916-related mobile element. *Plasmid* **58**: 61–67.
- Roberts AP, Johanesen PA, Lyras D, Mullany P & Rood JI (2001) Comparison of Tn5397 from *Clostridium difficile*, Tn916 from *Enterococcus faecalis* and the CW459tet(M) element from *Clostridium perfringens* shows that they have similar conjugation regions but different insertion and excision modules. *Microbiology* **147**: 1243–1251.
- Rocco JM & Churchward G (2006) The integrase of the conjugative transposon Tn916 directs strand- and sequence-specific cleavage of the origin of conjugal transfer, *oriT*, by the endonuclease Orf20. *J Bacteriol* **188**: 2207–2213.
- Romo M, Goicoechea P, Coque TM, Baquero F, de la Cruz F, Martínez-Martínez L & Francia MV (2008) *Molecular Typing of the Resistance Plasmids of a VRE Nosocomial Outbreak in Spain*. 18th European Congress of Clinical Microbiology and Infectious Diseases. April 2008. Barcelona, Spain.
- Rotger R & Casades J (1999) The virulence plasmids of *Salmonella*. *Int Microbiol* **2**: 177–184.
- Salgado-Pabon W, Jain S, Turner N, van der Does C & Dillard JP (2007) A novel relaxase homologue is involved in chromosomal DNA processing for type IV secretion in *Neisseria gonorrhoeae*. *Mol Microbiol* **66**: 930–947.
- Scheiffele P, Pansegrau W & Lanka E (1995) Initiation of *Agrobacterium tumefaciens* T-DNA processing. Purified proteins VirD1 and VirD2 catalyze site- and strand-specific cleavage of superhelical T-border DNA *in vitro*. *J Biol Chem* **270**: 1269–1276.
- Schneiker S, Keller M, Droge M, Lanka E, Puhler A & Selbitschka W (2001) The genetic organization and evolution of the broad host range mercury resistance plasmid pSB102 isolated from a microbial population residing in the rhizosphere of alfalfa. *Nucleic Acids Res* **29**: 5169–5181.
- Schubert S, Dufke S, Sorsa J & Heesemann J (2004) A novel integrative and conjugative element (ICE) of *Escherichia coli*: the putative progenitor of the *Yersinia* high-pathogenicity island. *Mol Microbiol* **51**: 837–848.
- Shen P, Jiang Y, Zhou Z, Zhang J, Yu Y & Li L (2008) Complete nucleotide sequence of pKP96, a 67850 bp multiresistance plasmid encoding *qnrA1*, *aac(6′)-Ib-cr* and *blaCTX-M-24* from *Klebsiella pneumoniae*. *J Antimicrob Chemother* **62**: 1252–1256.
- Sherburne CK, Lawley TD, Gilmour MW et al. (2000) The complete DNA sequence and analysis of R27, a large IncHI plasmid from *Salmonella typhi* that is temperature sensitive for transfer. *Nucleic Acids Res* **28**: 2177–2186.
- Shintani M, Nojiri H, Yoshida T, Habe H & Omori T (2003) Carbazole/dioxin-degrading *car* gene cluster is located on the chromosome of *Pseudomonas stutzeri* strain OM1 in a form different from the simple transposition of Tn4676. *Biotechnol Lett* **25**: 1255–1261.
- Shon M, Germino J & Bastia D (1982) The nucleotide sequence of the replication origin *beta* of the plasmid R6K. *J Biol Chem* **257**: 13823–13827.
- Smith CJ & Parker AC (1998) The transfer origin for *Bacteroides* mobilizable transposon Tn4555 is related to a plasmid family from gram-positive bacteria. *J Bacteriol* **180**: 435–439.
- Strauch E, Goelz G, Knabner D, Konietzny A, Lanka E & Appel B (2003) A cryptic plasmid of *Yersinia enterocolitica* encodes a conjugative transfer system related to the regions of CloDF13 Mob and IncX Pil. *Microbiology* **149**: 2829–2845.
- Sziper CY, Faelen M & Couturier M (2001) Mobilization function of the pBHR1 plasmid, a derivative of the broad-host-range plasmid pBBR1. *J Bacteriol* **183**: 2101–2110.
- Tauch A, Schneiker S, Selbitschka W et al. (2002) The complete nucleotide sequence and environmental distribution of the cryptic, conjugative, broad-host-range plasmid pIPO2 isolated from bacteria of the wheat rhizosphere. *Microbiology* **148**: 1637–1653.

- Thompson JD, Higgins DG & Gibson TJ (1994) CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Res* **22**: 4673–4680.
- Thorsted PB, Macartney DP, Akhtar P *et al.* (1998) Complete sequence of the IncPbeta plasmid R751: implications for evolution and organisation of the IncP backbone. *J Mol Biol* **282**: 969–990.
- Tomita H, Pierson C, Lim SK, Clewell DB & Ike Y (2002) Possible connection between a widely disseminated conjugative gentamicin resistance (pMG1-like) plasmid and the emergence of vancomycin resistance in *Enterococcus faecium*. *J Clin Microbiol* **40**: 3326–3333.
- Tomita H, Tanimoto K, Hayakawa S, Morinaga K, Ezaki K, Oshima H & Ike Y (2003) Highly conjugative pMG1-like plasmids carrying Tn1546-like transposons that encode vancomycin resistance in *Enterococcus faecium*. *J Bacteriol* **185**: 7024–7028.
- Tschape H & Tietze E (1980) Genetic and molecular characterization of R plasmids incompatible with R387 (IncK). *J Gen Microbiol* **118**: 515–521.
- van der Meer JR & Sentchilo V (2003) Genomic islands and the evolution of catabolic pathways in bacteria. *Curr Opin Biotechnol* **14**: 248–254.
- van Zyl LJ, Deane SM & Rawlings DE (2003) Analysis of the mobilization region of the broad-host-range IncQ-like plasmid pTC-F14 and its ability to interact with a related plasmid, pTF-FC2. *J Bacteriol* **185**: 6104–6111.
- Varsaki A, Lucas M, Afendra AS, Drinas C & de la Cruz F (2003) Genetic and biochemical characterization of MbeA, the relaxase involved in plasmid ColE1 conjugative mobilization. *Mol Microbiol* **48**: 481–493.
- Vedantam G, Knopf S & Hecht DW (2006) *Bacteroides fragilis* mobilizable transposon Tn5520 requires a 71 base pair origin of transfer sequence and a single mobilization protein for relaxosome formation during conjugation. *Mol Microbiol* **59**: 288–300.
- Villa L, Pezzella C, Tosini F, Visca P, Petrucca A & Carattoli A (2000) Multiple-antibiotic resistance mediated by structurally related IncL/M plasmids carrying an extended-spectrum beta-lactamase gene and a class 1 integron. *Antimicrob Agents Ch* **44**: 2911–2914.
- Waldor MK, Tschape H & Mekalanos JJ (1996) A new type of conjugative transposon encodes resistance to sulfamethoxazole, trimethoprim, and streptomycin in *Vibrio cholerae* O139. *J Bacteriol* **178**: 4157–4165.
- Welch TJ, Fricke WF, McDermott PF *et al.* (2007) Multiple antimicrobial resistance in plague: an emerging public health risk. *PLoS ONE* **2**: e309.
- Woese CR, Kandler O & Wheelis ML (1990) Towards a natural system of organisms: proposal for the domains Archaea, Bacteria, and Eucarya. *P Natl Acad Sci USA* **87**: 4576–4579.
- Womble DD & Rownd RH (1988) Genetic and physical map of plasmid NR1: comparison with other IncFII antibiotic resistance plasmids. *Microbiol Rev* **52**: 433–451.
- Zambrisky P, Tempe J & Schell J (1989) Transfer and function of T-DNA genes from agrobacterium Ti and Ri plasmids in plants. *Cell* **56**: 193–201.
- Zou X, Caufield PW, Li Y & Qi F (2001) Complete nucleotide sequence and characterization of pUA140, a cryptic plasmid from *Streptococcus mutans*. *Plasmid* **46**: 77–85.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. List of MOB_F plasmids^a.

Table S2. List of MOB_H plasmids^a.

Table S3. List of MOB_C plasmids^a.

Table S4. List of MOB_Q plasmids^a.

Table S5. List of MOB_P plasmids^a.

Table S6. List of MOB_{HEN} plasmids^a.

Table S7. List of MOB_V plasmids^a.

Fig. S1. Phylogeny tree of MOB_H T4CP.

Fig. S2. Phylogeny tree of MOB_C T4CP.

Fig. S3. Phylogeny tree of MOB_Q T4CP.

Fig. S4. Phylogeny tree of MOB_{P1} T4CP.

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